

Concept-anchored foundation model enables auditable and generalizable disease detection from routine chest CT

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ABSTRACT

While computed tomography (CT) is indispensable for diagnosis, staging, risk stratification and opportunistic screening, current foundation AI models for CT remain difficult to trust because their predictions are typically mediated by opaque latent representations and often degrade on new cohorts without clear explanation. This contrasts sharply with radiological practice, in which clinicians interpret CT scans by referencing a shared vocabulary of anatomical structures, imaging findings and disease-associated patterns. We introduce TRACE, a concept-anchored CT foundation model that learns auditable visual representations from routine volumetric CT and radiology reports. TRACE is developed using 3.9 million CT slices and 24,000 radiology reports through task-agnostic pretraining that aligns CT volumes and radiological reports. Each CT examination is represented not only by a visual embedding but also by a ranked profile of clinically readable concepts, allowing model predictions to be traced back to specific imaging evidence. Evaluated across CT-RATE, RadChestCT, Merlin, and INSPECT, TRACE can be transferred to diverse downstream tasks involving conventional thoracic findings, anatomy-derived labels, and EHR-derived opportunistic phenotypes, achieving competitive or superior performance in both zero-shot inference and supervised linear probing compared with strong CT-native and general vision-language baselines. Beyond disease detection, TRACE enables systematic concept auditing by recovering expected diagnostic evidence, exposing confounding and artefactual shortcuts, and revealing clinician-recognized, hypothesis-generating imaging-disease associations beyond manually predefined concept lists. TRACE also supports agentic concept refinement, improving the clinical relevance and interpretability of exposed concepts while maintaining predictive performance. TRACE represents a practical step toward accurate, generalizable, and inspectable CT modelling, with the potential to support opportunistic screening, and potential biomarker discovery.

1 Introduction

Computed tomography (CT) is one of the most widely used imaging modalities in modern medicine, providing rapid, high-resolution, and volumetric assessment across a broad range of diagnostic indications^{1,2}. Recent advances in foundation AI model have led to substantial progress in CT analysis, including semantic segmentation³⁻⁹, disease classification and multimodal prognosis¹⁰⁻¹⁴, radiology report generation¹⁵⁻¹⁸, and generalist vision-language modeling^{2,19}. In parallel, routine CT is

emerging as a powerful substrate for opportunistic screening². A scan acquired for one indication can also encode biomarkers of bone density, body composition, vascular calcification, cardiac morphology, liver steatosis and fibrosis, malignancy, and future cardiometabolic risk²⁰⁻²³. These signals have supported automated prediction of cardiovascular events and mortality from abdominal body composition²³, coronary artery calcium assessment from non-gated chest CT²⁴, cardiac and coronary calcium quantification for mortality prediction²⁵, opportunistic detection and staging of steatotic liver disease^{26,27}, and tumor detection from non-contrast CT²⁸⁻³⁵. Yet this promise is tempered by a central limitation of current CT foundation models: they rely on opaque latent representations that are difficult to audit, particularly in opportunistic prediction^{2,36}, where the target phenotype is often not the reason for imaging, the relevant evidence may be subtle or distributed across organs, and routine scans vary substantially across scanners, reconstruction kernels, contrast phases, anatomical coverage, and patient populations. These factors create many opportunities for models to exploit shortcut correlations, acquisition artefacts, protocol differences, demographic proxies, or confounding comorbidities rather than disease-relevant signal^{2,37,38}. Consequently, the generalization of even high-performing models must often be taken on trust, with limited insight into the clinical concepts that drive predictions or the causes of performance degradation across deployment settings.

The broader machine learning community has made rapid progress in vision-language foundation models (VLFM)³⁹⁻⁴³, where large-scale image-text pretraining enables transferable representations for downstream classification, retrieval, and zero-shot recognition^{40,44-46}. In biomedical imaging, this paradigm is particularly appealing because clinical knowledge is communicated through natural language, including radiology reports, structured findings, and disease descriptions⁴⁷⁻⁵². Nevertheless, most VLFM still make predictions through high-dimensional embeddings rather than explicit medical concepts, making their decisions difficult to audit even when text supervision is used. Post-hoc explanations, such as saliency maps and gradient-based attribution⁵³⁻⁵⁶, can summarize image regions after prediction but do not ensure grounding in clinically meaningful evidence and may be unstable under distribution shift^{57,58}. Concept bottleneck models (CBMs) provide a more direct route to interpretability by forcing predictions to pass through human-understandable concepts^{59,60}; however, existing CBMs and related concept-based methods typically require pre-specified concept sets, dense concept annotations or task-specific concept engineering^{57,59,60}. These requirements limit their scalability to volumetric CT and make them poorly suited to opportunistic AI, where relevant concepts may be incompletely known, indirectly associated with the phenotype, distributed across anatomical regions, or absent from the original report^{59,61-63}. Generalization remains a further challenge for VLFM that trained on reports from one or a few institutions can inherit local reporting styles, imaging protocols, patient populations and disease spectra, making them vulnerable to shortcut learning and distribution shifts that degrade performance and fairness in external cohorts⁶⁴⁻⁶⁶. This is especially limiting for opportunistic screening, where EHR-derived phenotypes extend beyond the report-explicit findings represented in any single institutional corpus.

In this study, we introduce TRACE (Transparent and Auditable clinical Concept Embedding), a concept-grounded framework for volumetric CT that shifts opportunistic screening from opaque risk prediction toward auditable, concept-based phenotyping. TRACE addresses the limitations of report-specific supervision by using large-language-model (LLM) embeddings to organize radiological observations, disease descriptions and phenotype-relevant concepts in a shared semantic space. Rather than treating each institutional report phrase or EHR label as an isolated target, TRACE aligns semantically related concepts even when they differ in wording, granularity or source. TRACE first curates a CT concept space from free-text CT reports by extracting clinical observations and consolidating them into a refined concept bank (Figure 1a). Its visual encoder is initialized from DINOv3⁶⁷ and adapted to volumetric CT through attentive aggregation^{68,69} of slice-level visual tokens, allowing two-dimensional visual representations to capture three-dimensional anatomical context. TRACE is then pretrained on paired CT volumes and reports from CT-RATE¹⁹ using contrastive image-text learning, and evaluated across CT-RATE¹⁹, RAD-ChestCT¹⁰, Merlin² and INSPECT⁷⁰ cohorts with heterogeneous CT appearance (Figure 2a). At inference, TRACE assigns each CT volume similarity scores to concepts in the CT concept space and projects these scores through the corresponding large-language-model concept embeddings⁷¹⁻⁷³ to obtain a transparent representation of the examination (Figure 1e). Compared with conventional CBMs^{57,59,60}, this LLM-grounded concept bottleneck does not require dense manual concept labels or a small task-specific vocabulary; instead, it represents each CT through thousands of clinically recognizable observations and allows downstream linear probes to determine which concepts are predictive for each phenotype. TRACE can therefore support both conventional radiological classification and opportunistic phenotype prediction (Figure 1c-f, Figure 2a), while enabling retrospective concept-level auditing to identify disease-relevant concepts, confounders, artifacts and candidate imaging biomarkers (Figure 3).

2 Results

2.1 TRACE provides concept-grounded CT representations for zero-shot and supervised transfer

TRACE was designed to transform CT prediction from an opaque image-text matching problem into a concept-indexed reasoning process. Instead of representing a CT volume only as a latent vector, TRACE first maps each study into a frozen concept space comprising 97,280 radiological observations extracted from CT reports. Each prediction can therefore be

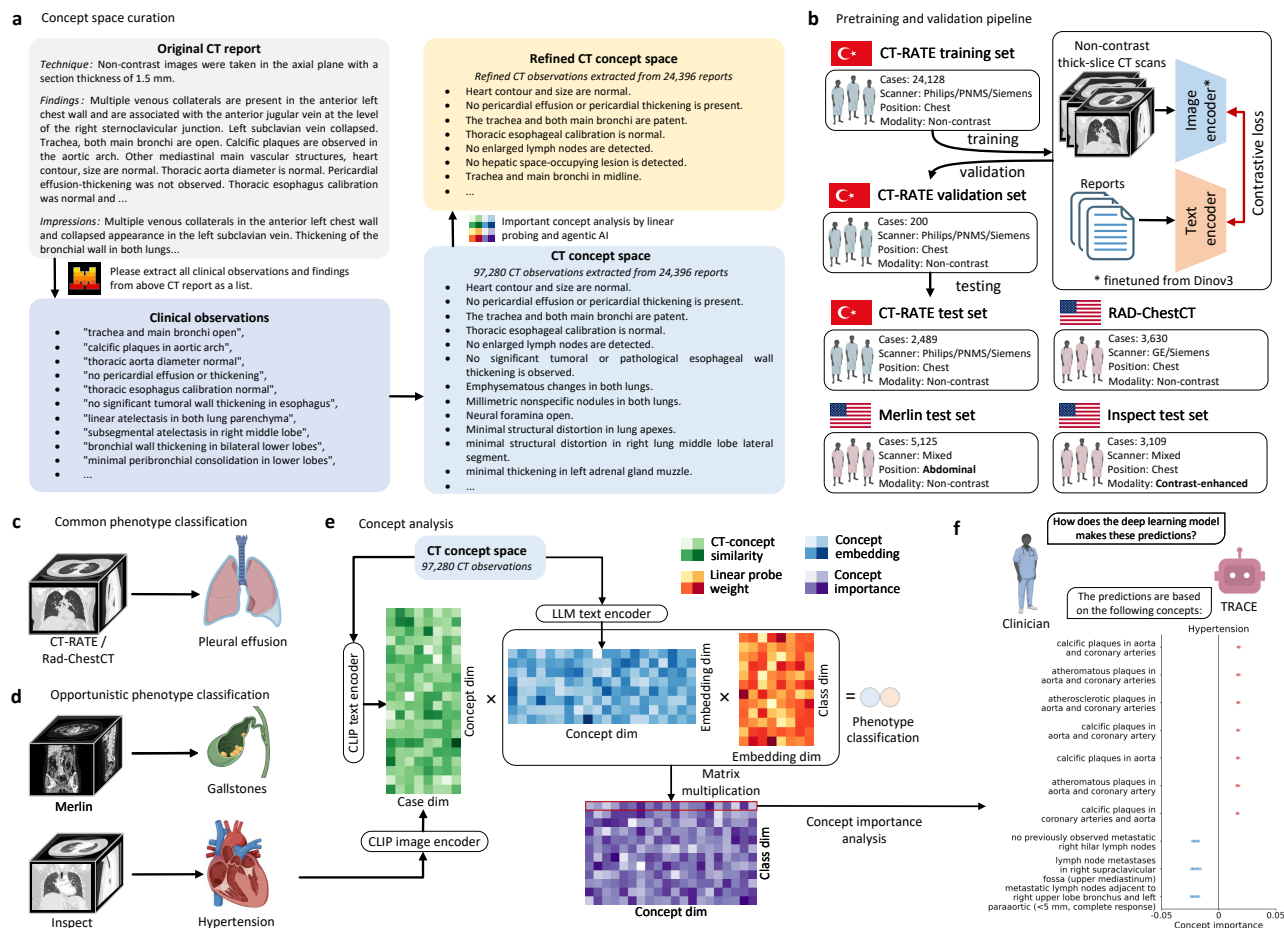


Figure 1. TRACE constructs auditable concept-level embeddings for volumetric CT. a, Concept space construction and refinement. Candidate CT concepts are mined from free-text radiology reports and consolidated into a refined concept bank that preserves clinically meaningful granularity while removing redundant or weakly informative observations. **b,** Pretraining, validation, and testing pipeline. TRACE is trained on paired CT volumes and radiology reports and validated across held-out CT cohorts, aligning volumetric image features with report-derived language representations through contrastive image-text learning. **c,d,** Reuse of the same concept-grounded representation for conventional and opportunistic CT tasks. TRACE supports common radiological phenotype classification and opportunistic phenotype prediction from routine CT examinations. **e,** Concept-level analysis for opportunistic prediction. For each CT volume, TRACE computes CT-concept similarities and combines them with large-language-model embeddings of the same concepts to generate transparent case-level embeddings. Linear probing on opportunistic tasks enables retrospective dataset-level analysis, separating canonical disease-related concepts, confounding concepts, and artefactual concepts, while also highlighting candidate concepts that may correspond to previously under-recognized CT findings associated with a phenotype. **f,** Transparent clinical interpretation and biomarker discovery. Downstream predictions can be decomposed into ranked CT concepts, allowing clinicians to inspect the evidence used by the model, audit potential shortcuts and generate hypotheses for new imaging biomarkers.

accompanied by a ranked profile of clinically readable CT concepts. This design builds on the empirical success of contrastive vision-language pretraining and recent 3D CT foundation models, while addressing a limitation that is particularly important in radiology: accurate models are not necessarily auditable unless their evidence can be expressed in the language used by clinicians^{2,54,74,75}. For zero-shot transfer, TRACE compares the concept activation profile of each CT volume with positive and negative language prompts for each disease label, producing disease scores without task-specific training (Fig. 2a). For supervised transfer, the same frozen TRACE features are passed to a lightweight linear classifier, allowing labelled data to recalibrate decision boundaries while preserving the concept basis on which each prediction rests (Fig. 2g). We evaluated this formulation across CT-RATE, RadChestCT, Merlin chest-part labels, Merlin opportunistic labels and INSPECT, spanning conventional thoracic findings, anatomy-derived labels and EHR-derived opportunistic phenotypes from CTPA^{2,70}. The detailed implementations are available in the Methods section.

In the zero-shot setting, TRACE demonstrated that a concept-mediated representation can retain competitive performance while making the evidence for each prediction directly inspectable. On CT-RATE, TRACE achieved a macro AUROC of 0.731 across 18 findings, comparable to CT-CLIP (0.727) and substantially above Merlin (0.591), MedSigLIP (0.620), BiomedCLIP (0.523) and OpenAI CLIP (0.492). TRACE was the top-performing method for 8 of 18 labels, including pleural effusion (AUROC, 0.921), arterial wall calcification (0.856), coronary artery wall calcification (0.845), cardiomegaly (0.824) and consolidation (0.824) (Fig. 2b; Figs. S1-S5). On RadChestCT, TRACE achieved the highest macro AUROC among all compared methods (0.653) and was best for 15 of 18 findings, despite differences in report style, label extraction and disease prevalence across the source cohorts. On Merlin, TRACE remained competitive across the combined chest-part and opportunistic label set (macro AUROC, 0.633), approaching the Merlin model itself (0.656) and exceeding MedSigLIP, CT-CLIP, BiomedCLIP and OpenAI CLIP in aggregate. On INSPECT, the largest zero-shot evaluation in our study with 3,109 CTPA studies and 97 EHR-derived phenotypes, TRACE reached up to 0.628 macro AUROC across language encoders, exceeding direct CLIP matching (0.577) and other foundation models. The radial heatmap in Fig. 2a further shows that this performance was distributed across respiratory, cardiovascular, gastrointestinal, neurological and systemic phenotypes rather than being confined to a narrow set of visually obvious labels. Full class-wise estimates, positive counts and 95% confidence intervals are reported in Tables S1, S2, S3, S4 and S5. These results indicate that CT concept embedding activations can function as a transferable semantic interface between volumetric imaging and disease labels, analogous in spirit to language-image pretraining but with the intermediate representation made explicitly auditable^{46,74,76}.

Linear probing (LP) tested a complementary question: whether the same transparent concept basis contains label-relevant signal that can be extracted with modest supervision. Using frozen TRACE concept-anchored features, a linear classifier achieved a macro AUROC of 0.809 on CT-RATE, outperforming the strongest non-TRACE vision-language baseline in the radial comparison (CT-CLIP, 0.741). TRACE was the best method for all 18 CT-RATE findings (Fig. 2g; Figs. S6-S8). On INSPECT, TRACE reached a macro AUROC of 0.689 across 97 phenotypes, compared with 0.634 for DINOv3 linear probing and 0.622 for CBM, with strong performance for pleurisy or pleural effusion (0.841), respiratory cancer (0.838), empyema and pneumothorax (0.824), pulmonary inflammation or oedema (0.822), ascites (0.821) and pneumonia (0.818). This comparison is important because DINO-style self-supervised vision encoders are strong general feature extractors, and CBMs provide a natural interpretable baseline in which predictions are mediated by concepts^{67,77,78}. The linear-probing ablation study in Fig. S1g compares direct visual features with concept-grounded representations. DINOv3 LP uses image features directly, whereas CBM and TRACE operate on the full LLM-derived concept set. On the in-distribution CT-RATE test set, TRACE achieved the highest macro-AUROC (0.809), modestly improving over DINOv3 LP (0.792) and CBM (0.774) while retaining an interpretable concept interface. On the out-of-distribution INSPECT cohort, the gain was larger: TRACE reached 0.689 macro-AUROC, outperforming DINOv3 LP by 5.5% and CBM by 6.7%. These results suggest that LLM-grounded concept representations enable more effective classification of broader EHR-derived phenotypes without retraining the CLIP backbone, while providing substantially lower variability than CBM.

The case-level audit trail produced by TRACE is illustrated in Fig. S2. Heatmaps were generated using an cross-token similarity-based visualization strategy⁷⁹ that combines slice-level attention with within-slice patch attention to localize model evidence in volumetric images. In Fig. S2a, for a lung-opacity case, the highest-ranked concepts captured multilobar, peripheral and subpleural ground-glass opacities, crazy-paving pattern and COVID-19-compatible consolidation, with heat maps localizing the corresponding parenchymal abnormalities across informative slices. These terms match established thoracic-imaging descriptors for ground-glass opacity, consolidation and interstitial patterns^{80,81}. In Fig. S2b, for a pneumonia-related case, TRACE highlighted band-shaped soft-tissue densities, linear subsegmental atelectasis and lower-lobe consolidation, separating visually grounded atelectatic and infectious patterns that would otherwise be collapsed into an opaque classifier score⁸². In Fig. S2c, TRACE ranked an infrarenal abdominal aortic aneurysm and calcified atherosclerotic changes in the aorta and iliac or visceral branches, despite ground-truth labels emphasizing nonspecific symptoms and pulmonary diagnoses. Together, these examples show that TRACE links ranked radiological statements to image locations, enabling case-level auditing of supporting evidence and potential label incompleteness beyond conventional saliency maps, which highlight regions but do not specify the

clinical observation used by the model^{54,56,83}.

2.2 TRACE separates diagnostic evidence from confounding and reveals candidate imaging-disease associations

We next examined whether the full TRACE concept space could support retrospective interpretability as a clinical audit instrument, rather than serving only as an intermediate representation for prediction. To construct the audit, report-derived CT-RATE concepts were embedded with an LLM, linear probes were trained for INSPECT EHR phenotypes in the TRACE concept space, and disease-specific probe weights were projected back onto the concept bank to assign an importance score to each natural-language concept (Fig. 3a). This procedure produced, for each phenotype, a ranked concept profile that could be directly inspected by clinicians. Expert readers annotated the top-ranked concepts into four predefined categories: **expected evidence**, defined as known, specific and strongly supportive imaging evidence for the target disease; **confounding evidence**, defined as non-specific or correlative findings that may co-occur with the disease but are insufficient for diagnosis; **novel evidence**, defined as imaging information not currently considered part of the established disease definition but potentially relevant for hypothesis generation; and **artefactual evidence**, defined as negated, normal, template-derived or otherwise misleading concepts that should not be interpreted as positive diagnostic support. In this design, interpretability is treated not as a visual post hoc illustration, but as a structured audit of what the model has learned, separating canonical disease signs, plausible correlates, potential shortcuts and candidate imaging-disease associations. This framing follows the broader goal of concept-based modelling-to expose the intermediate variables connecting inputs to predictions-while extending it to a large CT vocabulary for which dense manual concept annotation would be impractical^{78,84–86}.

The resulting audit profiles showed that TRACE recovered established medical knowledge while exposing where EHR-derived supervision departed from direct radiological evidence. For image-defined phenotypes, the dominant concepts were often canonical: pleurisy or pleural effusion was driven almost entirely by measured pleural-effusion descriptors, and pneumonia emphasized consolidation, crazy-paving and ground-glass patterns, consistent with conventional chest CT reasoning (Fig. 3b,f)^{80,82}. Other phenotypes revealed clinically indirect structure that would be invisible from AUROC alone. Empyema or pneumothorax mixed plausible pleural findings with co-occurring abdominal fluid concepts and negated absence statements, suggesting that the probe partly used reporting context and comorbidity rather than a single disease-specific sign (Fig. 3d). Non-malignant ascites similarly contained direct ascites and liver-disease concepts, while the broader audit identified non-specific pulmonary nodules among high-ranked concepts, consistent with disease-burden confounding rather than a mechanistic sign of ascites (Fig. 3e). These findings reflect a central challenge in medical imaging AI: models may learn stable but clinically indirect correlations when labels are derived from reports, billing codes or heterogeneous clinical pathways^{38,75,87}. Importantly, the same audit also surfaced candidate imaging-disease associations. Respiratory cancer was associated not only with malignant descriptors but also with bronchiectasis, emphysematous-fibrotic and chronic lung-remodeling patterns; obesity-related phenotypes highlighted atelectasis and mosaic-attenuation concepts; and shock was accompanied by small subpleural nodules and ground-glass opacity patterns (Fig. 3c,h,i). We interpret these findings conservatively: they are not causal claims, but prioritized associations that can be tested through covariate adjustment, external validation and prospective clinical review. Thus, TRACE transforms a high-dimensional concept bank into an auditable map of established evidence, confounding structure, artefactual language and candidate discoveries. In Supplementary Fig. S3, clinician annotation of the top-ranked TRACE concepts shows that conventional CT-RATE phenotypes are dominated by expected radiological evidence, whereas INSPECT EHR-derived phenotypes exhibit a broader mixture of expected concepts, candidate associations, confounders and occasional artefacts. This contrast suggests that TRACE recovers direct imaging evidence for image-defined findings while exposing the more heterogeneous supervision signals underlying opportunistic EHR phenotypes. The class-wise audits in Figs. S4–S7 further show that physicians can inspect the top-10 concepts for each disease and determine whether TRACE relies on canonical signs, clinically plausible but non-specific correlates, artefactual report language or hypothesis-generating associations. Together, these updated disease-level profiles demonstrate that TRACE supports systematic clinician review of model reasoning across both standard CT findings and broad EHR-derived disease labels.

Finally, we asked whether the gain in interpretability and auditability came at the cost of diagnostic performance. In a reader study of 149 CT volumes, two radiologists with over 5 year experience independently assessed the selected EHR-derived phenotypes, and their readings were compared with TRACE predictions obtained after linear probing. TRACE achieved higher average performance than the mean radiologist reading across the four top-ranked INSPECT categories, with AUROC increasing from 0.615 to 0.789, MCC from 0.220 to 0.365 and F1 from 0.392 to 0.540 (Table 1). The largest gains were observed for pleural thickening and pleural calcification, where radiologist performance was limited but TRACE retained strong discrimination; radiologists remained competitive for visually salient or threshold-sensitive findings such as pleural effusion and ascites. These results suggest that TRACE can retain the advantages of concept-based modeling-transparent evidence tracing and discovery-oriented auditing-while achieving diagnostic performance that exceeds human readers for several EHR-linked phenotypes. The comparison should not be read as a claim of clinical deployment readiness. Rather, it demonstrates that

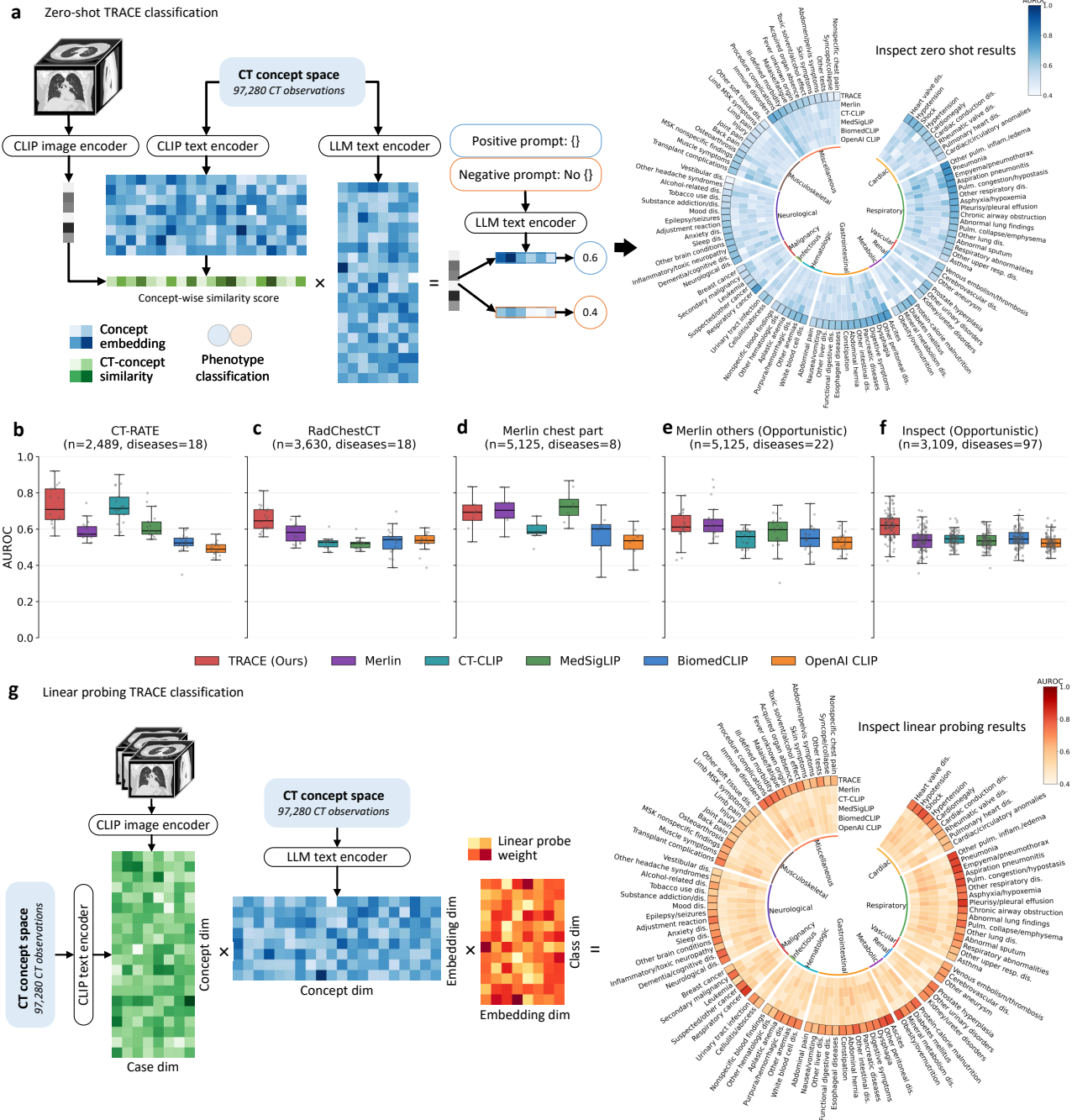


Figure 2. TRACE enables transparent zero-shot and supervised CT phenotype classification through a complete concept space. **a**, Zero-shot classification. TRACE projects each CT volume into a 97,280-dimensional clinical concept space and compares the resulting concept activation profile with positive and negative text prompts for each phenotype. The radial heatmap summarizes zero-shot AUROC on INSPECT across 97 EHR-derived opportunistic phenotypes. **b-f**, Zero-shot AUROC distributions across CT-RATE ($n = 2489$, 18 findings), RadChestCT ($n = 3630$, 18 findings), Merlin chest-part labels ($n = 5125$, 8 findings), Merlin opportunistic labels ($n = 5125$, 22 findings) and INSPECT opportunistic labels ($n = 3109$, 97 phenotypes). TRACE is compared with Merlin, CT-CLIP, MedSigLIP, BiomedCLIP and OpenAI CLIP; each point denotes one phenotype. TRACE achieves competitive or superior zero-shot performance while preserving an auditable prediction. **g**, Linear probing uses frozen TRACE concept features as input to train a linear classifier. The schematic illustrates supervised calibration with the concept-grounded representation, and the radial heatmap shows AUROC on INSPECT.

Table 1. Reader study on EHR-derived INSPECT phenotypes. TRACE was compared with junior and senior radiologists on $n = 149$ CT volumes with INSPECT linear-probe scores. The four findings are the INSPECT categories with the highest TRACE AUROC in this cohort. Radiologist performance is averaged across readers, and TRACE denotes the TRACE-CBM checkpoint with the highest validation AUROC. Estimates are shown with 95% patient-level bootstrap confidence intervals; p values are two-sided bootstrap tests for $\Delta = \text{TRACE} - \text{radiologists}$. **Bold** marks the best result for finding.

	Average	Pleural effusion	Pleural thickening	Pleural calcification	Ascites
<i>AUROC</i>					
Radiologists (mean)	0.615 (0.585, 0.650)	0.765 (0.708, 0.823)	0.523 (0.495, 0.554)	0.495 (0.488, 0.500)	0.676 (0.593, 0.785)
TRACE	0.789 (0.731, 0.852)	0.802 (0.737, 0.871)	0.801 (0.737, 0.870)	0.785 (0.711, 0.850)	0.767 (0.644, 0.869)
Δ (TRACE - Radiologists)	0.174 (0.115, 0.229)	0.037 (-0.019, 0.098)	0.278 (0.210, 0.346)	0.290 (0.216, 0.355)	0.091 (-0.045, 0.211)
p -value	< 0.001	0.220	< 0.001	< 0.001	0.175
<i>MCC</i>					
Radiologists (mean)	0.220 (0.143, 0.294)	0.499 (0.394, 0.612)	0.069 (-0.018, 0.170)	-0.058 (-0.088, 0.000)	0.369 (0.159, 0.567)
TRACE	0.365 (0.287, 0.442)	0.414 (0.308, 0.522)	0.414 (0.308, 0.522)	0.375 (0.263, 0.474)	0.258 (0.119, 0.374)
Δ (TRACE - Radiologists)	0.146 (0.069, 0.231)	-0.085 (-0.195, 0.032)	0.346 (0.225, 0.460)	0.433 (0.298, 0.528)	-0.111 (-0.333, 0.138)
p -value	< 0.001	0.150	< 0.001	< 0.001	0.375
<i>F1 score</i>					
Radiologists (mean)	0.392 (0.319, 0.451)	0.677 (0.594, 0.758)	0.308 (0.243, 0.383)	0.000 (0.000, 0.000)	0.584 (0.373, 0.745)
TRACE	0.540 (0.462, 0.613)	0.621 (0.530, 0.716)	0.621 (0.530, 0.716)	0.610 (0.521, 0.692)	0.308 (0.181, 0.434)
Δ (TRACE - Radiologists)	0.147 (0.085, 0.202)	-0.057 (-0.121, 0.010)	0.312 (0.241, 0.387)	0.610 (0.521, 0.692)	-0.277 (-0.425, -0.113)
p -value	< 0.001	0.105	< 0.001	< 0.001	< 0.001

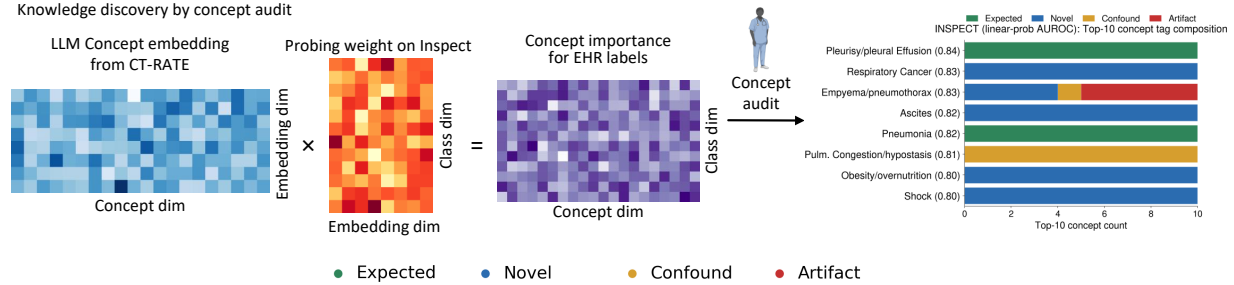
concept-level supervision can support prediction and knowledge discovery in the same model, provided that the output remains subject to expert review, external validation and clinically meaningful operating-point selection^{88–90}.

2.3 TRACE enables agentic concept audit and refinement

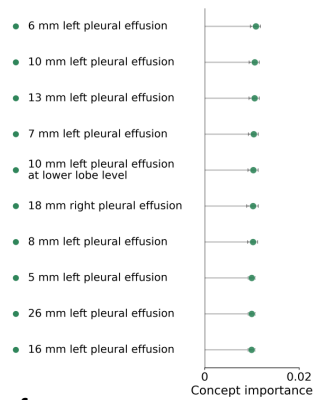
A central feature of TRACE is that its prediction layer expressed in a language space that can be inspected, scored and edited. We therefore evaluated whether radiological concepts selected by TRACE constituted meaningful evidence for each target disease. For this purpose, LLMs were used as scalable concept judges, assigning each concept-disease pair an integer relevance score from 1 to 5, ranging from unrelated or potentially misleading evidence to directly diagnostic evidence. In parallel, each concept was assigned to one of four audit categories mentioned before: *expected* concepts, representing established and relatively specific disease evidence; *confounder* concepts, representing known but non-specific supportive findings; *novel* concepts, representing candidate disease-associated information not currently regarded as established for that label; and *artifact* concepts, representing exclusionary, healthy or misleading evidence. This use of language models as scalable semantic judges follows the broader development of text-embedding and language-guided concept methods, but we anchored the process to blinded clinical review because language-model plausibility alone is insufficient for medical validation^{84, 85, 91, 92}. To ensure that this language-based audit was not an unconstrained post hoc annotation, we performed a blinded clinical validation on 100 sampled concept-disease pairs, which were independently scored by two clinical experts on CT-RATE and INSPECT report-derived concepts. LLM relevance scores showed moderate-to-strong Kendall agreement with the two expert raters across datasets and clinicians (Fig. 4a), supporting the use of LLMs as scalable concept judges while preserving expert assessment as the clinical reference standard.

We next investigated whether concept refinement could improve the concept interpretability without losing the classification accuracy. TRACE first estimates class-specific concept importance over the complete concept bank, then selects disease-relevant concepts and reuses this refined concept vocabulary for downstream concept analysis (Fig. S8). We examined how supervised linear-probe performance changed as the number of retained concepts varied from the top 100 to the full vocabulary (Fig. S9a). Across both CT-RATE and INSPECT, TRACE remained stable over a wide range of concept-bank sizes, indicating that the LLM-projected representation was not strongly dependent on an arbitrary top- K cutoff. By contrast, the conventional CBM⁵⁷ showed stronger dependence on concept-bank size: its performance increased as the concept space was reduced, especially when moving from the full concept bank to more compact top- K subsets. This pattern suggests that a standard CBM is more vulnerable to noisy, weakly related or redundant concepts when the vocabulary becomes large, whereas TRACE benefits from semantic aggregation in the LLM embedding space and can preserve predictive signal while suppressing irrelevant variation. We also observed that newer and larger language-embedding backends generally improved or maintained performance, with updated embedding models such as `text-embedding-3-large`, `voyage-3-large`, `voyage-4` and `voyage-4-large` performing among the strongest variants (Fig. S9a). The `voyage` models represent successive generations within the same model family, and their improved performance suggests that advances in newer and stronger

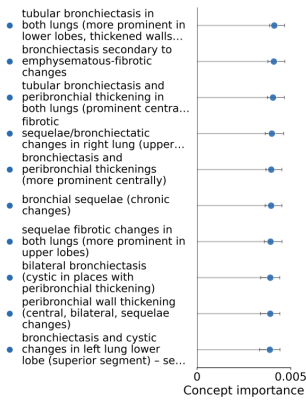
a Knowledge discovery by concept audit



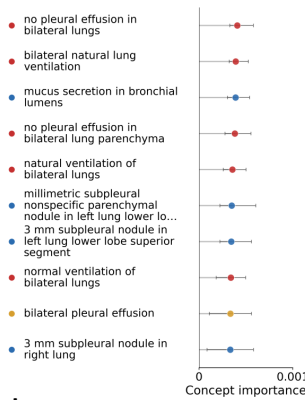
b Pleurisy/pleural effusion



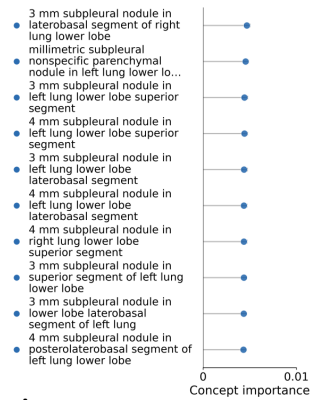
c Respiratory cancer



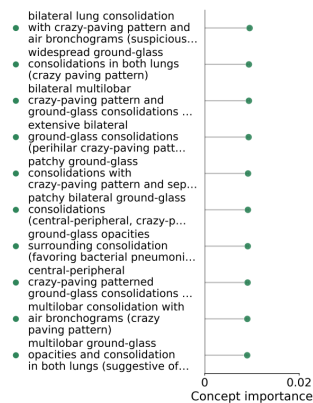
d Empyema/pneumothorax



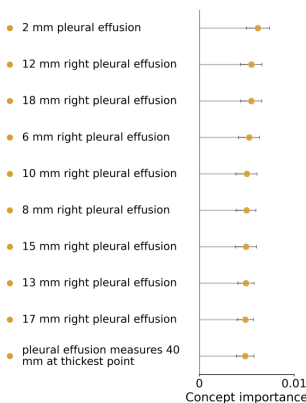
e Ascites



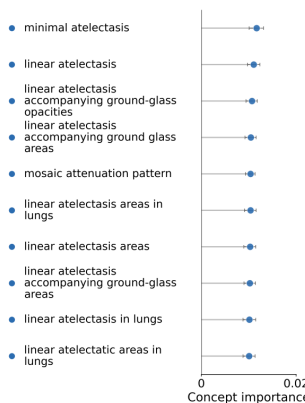
f Pneumonia



g Pulm. congestion/hypostasis



h Obesity/overnutrition



i Shock

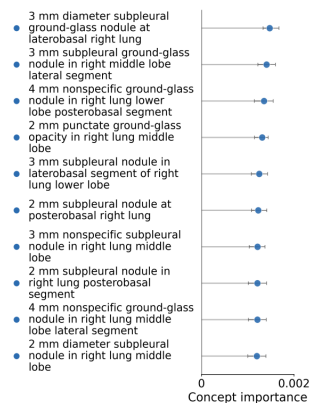


Figure 3. Concept-level auditing identifies diagnostic evidence, confounders, artefacts and hypothesis-generating disease associations. **a**, TRACE combines LLM-derived concept embeddings with disease-specific linear-probe weights to estimate class-level concept importance for INSPECT EHR phenotypes. Expert clinicians then assign the top-ranked concepts to four audit categories: expected diagnostic evidence, candidate novel association, confounder or artefact. **b-i**, Representative audits for high-performing INSPECT phenotypes. Pleurisy or pleural effusion is dominated by direct effusion descriptors (**b**); respiratory cancer highlights bronchiectatic, emphysematous-fibrotic and chronic lung-remodelling concepts that may represent candidate disease-associated imaging signals (**c**); empyema or pneumothorax contains a mixture of expected pleural findings, confounding co-occurrences and artefactual absence statements (**d**); non-malignant ascites contains ascites-related and liver-disease evidence together with non-specific pulmonary nodule concepts, suggesting that the model can expose both direct evidence and disease-burden confounding (**e**); pneumonia is supported by consolidation, crazy-paving and ground-glass concepts (**f**); pulmonary congestion or hypostasis is linked to pleural-effusion concepts (**g**); obesity or overnutrition highlights atelectatic and mosaic-attenuation patterns (**h**); and shock is associated with small subpleural nodules and ground-glass opacity patterns (**i**). Points denote concept-importance estimates, horizontal lines denote uncertainty intervals and colours denote clinician audit categories.

embedding models can translate directly into better TRACE performance^{92,93}. These results indicate that TRACE is not tied to a fixed language model, but can be continuously improved as general-purpose and domain-adapted embedding models evolve.

This refinement preserved predictive performance while substantially improving the clinical meaningfulness of the exposed concepts. On CT-RATE, TRACE achieved similar or higher AUROC than the CBM under both top-100 and full-concept settings, while producing markedly higher LLM explanation scores than the CBM (top-100: 2.97 versus 1.78; full: 2.84 versus 1.78; Fig. 4b). The class-level results further showed that TRACE improved interpretability without sacrificing discrimination. For example, in arterial wall calcification, TRACE with the refined concept set achieved an AUROC of 0.883 and an LLM relevance score of 4.60, whereas the CBM full-concept model achieved an AUROC of 0.857 and an LLM score of 1.10. In coronary artery wall calcification, TRACE achieved an AUROC of 0.875 with an LLM score of 4.40, compared with 0.852 and 1.35 for the CBM full-concept model (Tables S11-S10). Similar trends were observed on INSPECT. TRACE was performance-matched to the CBM in the top-100 setting, substantially outperformed the CBM in the full-concept setting (AUROC 0.689 versus 0.622), and produced more clinically meaningful explanations before and after refinement (top-100: 2.18 versus 1.57; full: 1.85 versus 1.57). At the phenotype level, TRACE yielded high-relevance explanations for clinically image-grounded labels, including pneumonia, chronic airway obstruction and abnormal lung findings. For instance, on INSPECT pneumonia, TRACE with the refined concept set achieved an AUROC of 0.818 and an LLM score of 3.60, whereas the CBM full-concept model achieved an AUROC of 0.722 and an LLM score of 1.15. For chronic airway obstruction, TRACE achieved an AUROC of 0.754 with an LLM score of 3.70, compared with 0.665 and 1.60 for the CBM full-concept model (Tables S11-S10). Thus, refinement did not simply compress the concept bank. Rather, it converted a large language-derived vocabulary into a disease-conditioned representation in which predictive performance was retained, conventional CBM failure modes were reduced and the concepts exposed to clinicians became more interpretable^{78,84,86}.

Qualitative inspection of representative phenotypes further showed that TRACE improved not only predictive performance but also the clinical specificity of the exposed concept profiles (Fig. S10). For arterial wall calcification (Fig. S10a) and coronary artery wall calcification (Fig. S10b), both TRACE variants prioritized anatomically matched vascular concepts, including calcific atheroma plaques in the ascending, thoracic and descending aorta, as well as the coronary arteries. By contrast, CBM more often surfaced less specific or anatomically displaced concepts, such as pleural retraction, biliary or portal venous findings, abdominal aortic abnormalities and heterogeneous abdominal-organ descriptions. For pneumonia (Fig. S10c) and abnormal lung findings (Fig. S10d), TRACE emphasized pulmonary parenchymal patterns that are directly aligned with thoracic CT interpretation, including bilateral multilobar consolidation, crazy-paving, ground-glass opacities and patchy air-space disease, whereas CBM mixed these signals with broader findings such as abdominal fluid, pacemaker-related terms or metastatic and postoperative descriptors. The same pattern was seen for chronic airway obstruction (Fig. S10e) and abnormal sputum (Fig. S10f): TRACE highlighted airway-centred findings, including bronchiectasis, peribronchial thickening, bronchial wall thickening and related obstructive changes, while CBM more frequently included diffuse or extra-pulmonary concepts, such as pleural collections, free abdominal fluid, vascular interventions and nonspecific abdominal abnormalities. Together, these examples show that TRACE produces disease-conditioned concept profiles that are more anatomically grounded, clinically coherent and easier to inspect than those of CBM.

Finally, we used LLM agents to turn concept auditing into a mechanism for targeted model editing. For a selected disease, the agent re-scored all concepts in the refined vocabulary and assigned each concept to the predefined audit categories under clinician supervision (Fig. 4c). The refined concept subset was then used to rerun TRACE on classes in which supervised linear probing was less favourable or clinically less informative, as illustrated by the INSPECT analysis of inflammatory or toxic neuropathy (Fig. S11). In this case, retaining concepts with higher LLM relevance scores improved AUROC across random seeds, with gains increasing as the relevance threshold became more stringent (Fig. 4d). Category-based editing provided a complementary audit. Retaining *expected* concepts improved performance, confirming that TRACE can recover established radiological evidence. Importantly, retaining *novel* concepts also produced a clear performance gain, and combining expected and novel concepts remained beneficial (Fig. 4e). These results provide an initial demonstration that concepts beyond current expert-recognized diagnostic knowledge may contain useful predictive signal. However, such concepts should be interpreted as candidate imaging biomarkers rather than established clinical knowledge. Their biological plausibility, reproducibility across cohorts, independence from confounding factors and added value over standard radiological criteria require further validation in dedicated clinical and mechanistic studies^{90,94,95}. Taken together, TRACE explanations are not static rationales appended to a classifier. They are editable clinical objects that can be audited, pruned and recomposed, exposing known disease evidence while also generating testable hypotheses for radiological knowledge discovery.

3 Discussion

Most existing CT foundation models and vision-language models have focused on learning transferable image-text representations from radiology reports^{2,49,74}, but have not made the intermediate clinical evidence used for prediction directly auditable. This leaves a substantial gap for routine CT^{1,96}, where examinations are volumetric, multi-organ and protocol-dependent,

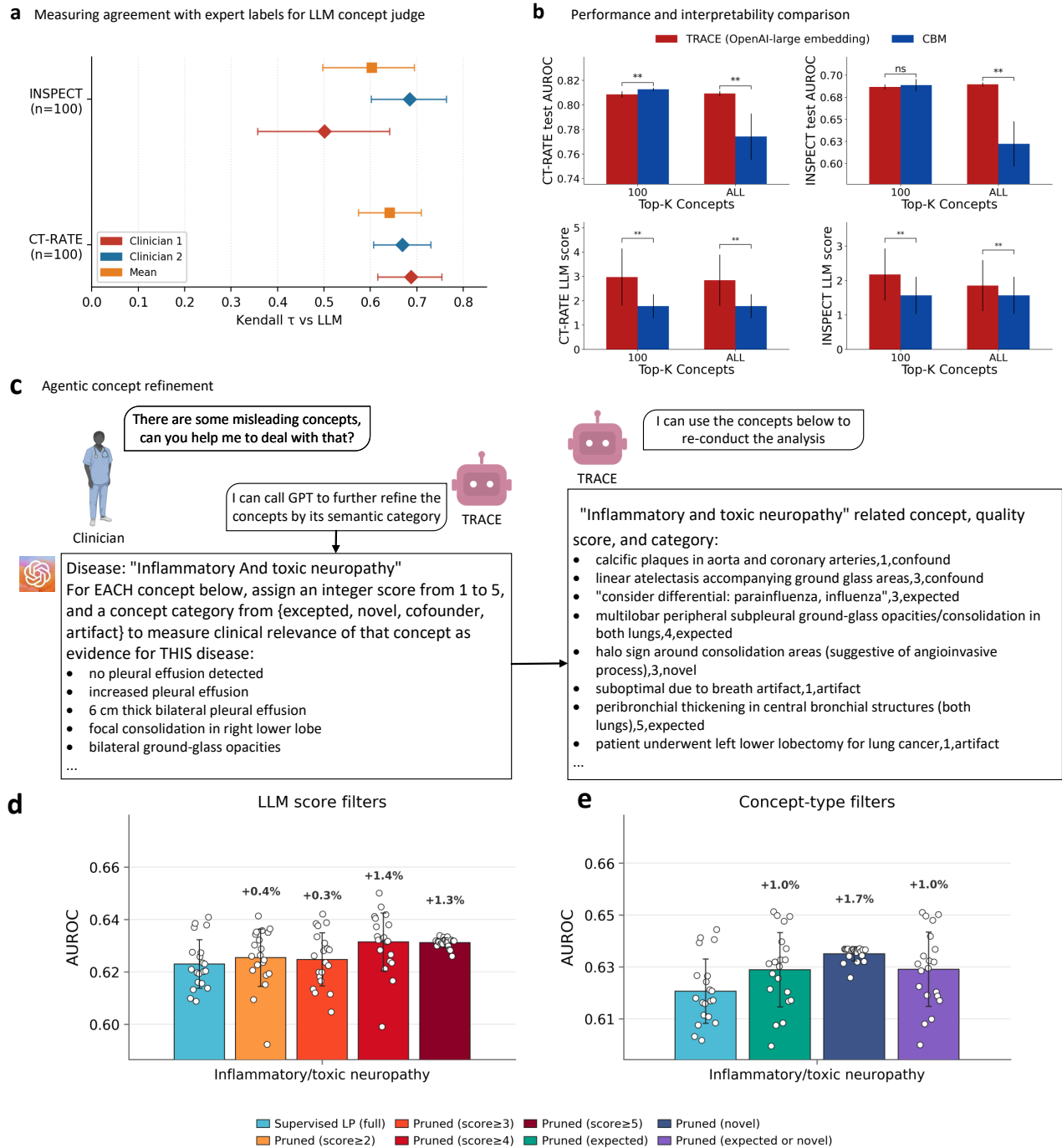


Figure 4. Concept refinement improves interpretability and enables agentic concept-space editing. **a**, Concordance between the LLM concept judge and blinded radiologist annotations for CT-RATE and INSPECT concept-disease pairs. Concepts were scored from 1 to 5 for clinical relevance; points show Kendall's τ and error bars show confidence intervals. **b**, AUROC and concept-level interpretability scores for TRACE and conventional CBM variants using top-100 or full concept banks on CT-RATE and INSPECT. TRACE maintains competitive predictive performance while yielding more clinically relevant concepts. **c**, Agentic concept refinement workflow. For a selected disease, an agent scores concept relevance and assigns concepts to predefined audit categories, allowing TRACE to remove artefacts or confounders, retain expected evidence or candidate novel associations. **d**, Relevance-based filtering for the INSPECT phenotype inflammatory or toxic neuropathy. Retaining concepts with higher LLM relevance improves AUROC across random seeds. **e**, Category-based filtering for the same phenotype. Expected concepts improve performance, and candidate novel concepts provide additional predictive signal. Bars show means across 20 random seeds, points denote individual runs and error bars denote variability across seeds.

and where downstream opportunistic phenotypes may differ from the original imaging indication. Prior studies in real-world medical-imaging generalization^{66,87,97–100} have shown that models can fail when scanner settings, patient populations, disease spectra or acquisition protocols change. In this study, we introduced TRACE, a concept-grounded foundation model for volumetric CT that represents each examination through clinically meaningful natural-language concepts, using a DINOv3-initialized visual backbone⁶⁷, CT-report contrastive pretraining on CT-RATE¹⁹ and large-language-model semantic embeddings^{71–73,91,92}. Rather than explaining a black-box model after prediction, TRACE exposes the radiological concepts through which transfer occurs.

A key distinction between TRACE and conventional CLIP-style CT models^{2,74} is that the concept space is part of the model interface. Standard vision-language pretraining^{39,40,74} supports zero-shot recognition and retrieval, but predictions are usually mediated by latent image-text similarity scores that are difficult to decompose clinically. TRACE first describes each CT volume by its similarity to a large bank of report-derived CT concepts and then projects this concept profile into an LLM semantic space^{71–73}. This design is particularly useful for opportunistic screening^{96,101,102}, where EHR-derived phenotypes⁷⁰ may be broader than the findings explicitly described in the original report. Across CT-RATE¹⁹, RadChestCT¹⁰, Merlin² and INSPECT⁷⁰, TRACE supported both zero-shot inference and supervised linear probing. The ablation experiments further showed that replacing TRACE with direct DINOv3 linear probing⁶⁷ or a conventional concept bottleneck model^{59,60,84,86} reduced performance, especially for out-of-distribution EHR phenotypes, suggesting that LLM-grounded concept representations provide a useful bridge between report-derived CT evidence and broader clinical labels.

TRACE also reframes interpretability in CT from visual explanation to structured clinical auditing. Saliency maps^{55,56}, attention visualizations⁷⁹ and general post hoc explanation methods^{54,83} can localize image regions, but they do not directly specify whether a model relied on diagnostic evidence, comorbidity, acquisition artefact, negated language or incidental correlates. This distinction is important because hidden stratification³⁸ and shortcut learning^{37,64,75} can produce apparently strong aggregate performance while masking clinically meaningful failure modes. TRACE instead decomposes each disease model into ranked natural-language concepts, allowing clinicians to audit model behaviour using the same type of language used in radiological practice. In our audit, top-ranked concepts were organized as expected evidence, confounding evidence, candidate novel evidence or artefactual evidence. This structured review makes the model's reasoning inspectable at the level of disease signs, plausible correlates, report artefacts and hypothesis-generating associations, extending the goal of concept-based modelling^{59,84–86} to a large CT vocabulary where dense manual concept annotation would be impractical.

The concept-audit results illustrate both the reliability and the caution required when using report- or EHR-derived supervision. For image-defined phenotypes, TRACE recovered established thoracic-imaging evidence, including pleural effusion descriptors, consolidation, ground-glass opacity and crazy-paving patterns^{80–82}. For broader EHR-derived phenotypes⁷⁰, the same audit exposed more heterogeneous supervision signals: some concepts reflected plausible comorbidity or disease severity, whereas others reflected artefactual language, negated findings or clinically indirect correlations. Such behaviour is consistent with known challenges in medical imaging AI, including site-dependent generalization⁸⁷, hidden subgroup failure³⁸, shortcut learning^{37,75} and fairness degradation under external distribution shift^{66,103}. Importantly, TRACE also surfaced clinician-recognized candidate associations that are not established diagnostic criteria. These associations should not be interpreted as causal claims, but because they are expressed as readable CT concepts, they can be tested through covariate adjustment, external validation, site-stratified analysis and prospective clinical review^{90,94,104}.

Several limitations remain. The TRACE concept bank is derived from radiology reports and therefore inherits reporting bias, variable terminology, under-reporting, negation, templated language and institution-specific phrasing^{105–107}. EHR-derived labels^{70,102} are noisy proxies for imaging-visible disease, so high performance may reflect treatment history, disease severity, comorbidity or label leakage rather than the target mechanism itself. Candidate novel concepts require validation in larger and more diverse cohorts, including assessment of acquisition protocol, scanner vendor, contrast phase, anatomical coverage, demographic subgroup and site effects^{66,97,98}. Although TRACE was evaluated across multiple CT datasets, further work is needed to test calibration, fairness, multilingual reporting, rare phenotypes and prospective deployment in clinical workflows^{94,95,104}. More broadly, TRACE suggests that radiology foundation models need not choose between performance and interpretability: by grounding CT representations in auditable clinical concepts, they can support prediction, failure analysis, model editing and hypothesis generation within a single transparent framework.

4 Methods

4.1 Study design and data sources

This study developed and evaluated TRACE, a concept-grounded vision-language foundation model for volumetric computed tomography (CT). TRACE was trained using paired CT volumes and free-text radiology reports and was evaluated on conventional radiological findings, external CT benchmarks, and EHR-derived opportunistic phenotypes. The overall workflow consisted of five stages: construction of a report-derived CT concept space, CT-report contrastive pretraining, zero-shot and supervised transfer evaluation, concept-level model auditing, and clinician-based assessment.

CT-RATE for CT-report pretraining and conventional phenotype evaluation. We used CT-RATE¹⁹, a large-scale chest CT dataset containing volumetric CT examinations, radiology reports, and multi-label radiological finding labels. CT-RATE was used for task-agnostic CT-report contrastive pretraining and for evaluation on conventional radiological phenotypes. We adopted the released patient-level training, validation, and test splits to avoid information leakage across phases. For supervised evaluation, labels were treated as binary indicators and the task was formulated as multi-label classification.

RAD-ChestCT and Merlin for external CT evaluation. We used RAD-ChestCT¹⁰, and Merlin² as external evaluation cohorts. RAD-ChestCT contains chest CT volumes with multiple abnormality labels derived from radiology reports. Merlin provides an additional CT vision-language benchmark and label space. These datasets were used only for evaluation and were processed into the same volume-level inference format as CT-RATE.

INSPECT for EHR-derived opportunistic phenotype evaluation. We used INSPECT⁷⁰, a contrast-enhanced CT pulmonary angiography cohort linked to electronic health record (EHR) data, to evaluate opportunistic phenotype prediction beyond report-defined CT labels by following Merlin². EHR phenotypes were derived from OMOP Common Data Model tables, including `procedure_occurrence`, `visit_occurrence`, `condition_occurrence`, `concept`, and `concept_relationship`. Diagnosis codes were mapped to ICD-9-CM and ICD-10-CM code strings, aggregated into PheCodes using PheTK, and converted into binary visit-level labels before assignment to the corresponding CT volume. We applied hierarchy-aware roll-up, enforced mutually exclusive PheCode exclusions where specified, and removed rare phenotypes with fewer than a prespecified number of positive examples. Train, validation, and test splits were fixed at the volume level by intersecting the filtered phenotype table with dataset-specific split files.

CT pre-processing. Each CT scan was first loaded as a 3D volume and reoriented to a canonical RAS orientation. When DICOM metadata were available, voxel intensities were converted to Hounsfield units using the corresponding rescale slope and intercept. Intensities were then clipped to the range $[-1000, 1000]$ HU. Each volume was resized with trilinear interpolation while preserving aspect ratio, center-padded to a fixed spatial size of $160 \times 224 \times 224$, and padded with air-density values. The clipped HU values were then linearly mapped to 8-bit image intensities:

$$\tilde{x}_{d,h,w} = \text{clip} \left(\frac{x_{d,h,w}/1000 + 1}{2} \times 255, 0, 255 \right), \quad (1)$$

where d indexes the axial slice and h, w index in-plane spatial coordinates. This pre-processing produced fixed-size axial CT stacks with a standardized intensity range while preserving the relative contrast induced by the Hounsfield-unit scale.

Ethics. All datasets used in this study are publicly available and de-identified according to their respective data-use agreements and governance procedures^{2,10,19,70}. No additional patient data were collected for this study.

4.2 Construction of a report-derived CT concept space

TRACE uses a large natural-language concept space as the interface between volumetric CT images and downstream disease labels. The concept space was designed to preserve radiological granularity while reducing redundant phrasing, allowing model outputs to be inspected as clinically readable observations rather than only as latent vectors.

Mining candidate concepts from radiology reports. We first extracted candidate CT concepts from free-text radiology reports. Candidate concepts were defined as short clinical observations that describe anatomy, imaging findings, disease-relevant patterns, measurements, devices, procedures, or normal/negative statements. Examples include pleural-effusion measurements, ground-glass opacities, aortic calcification, bronchiectasis, pericardial effusion, lymph-node enlargement, hepatic steatosis, and post-procedural findings. This report-derived strategy was used because radiology reports encode the vocabulary through which clinicians routinely communicate CT findings.

Consolidating redundant report phrases into a refined concept bank. Because free-text reports contain repeated, synonymous, and template-derived phrases, we consolidated candidate concepts into a refined concept bank. Redundant phrases were merged when they described the same radiological observation, whereas clinically meaningful differences in anatomy, laterality, measurement or disease pattern were preserved. The resulting concept bank retained thousands of natural-language CT observations and was used for both conventional CT phenotype analysis and EHR-derived opportunistic phenotype prediction.

Encoding concepts in both vision-aligned and language-semantic spaces. For each concept c_k , we computed two aligned representations. First, a vision-aligned text embedding e_k^{vis} was obtained using the text pathway learned during CT-report contrastive pretraining, enabling direct cosine comparison between CT image embeddings and concept text. Second, a language-semantic embedding e_k^{LLM} was computed with large-language-model embedding backbones to organize concepts in a broader natural-language space^{71–73,91,92}. To assess whether TRACE depended on a specific text-embedding model, we

evaluated multiple embedding families, including OpenAI-text-embedding-3-large, Gemini-2, Voyage-4-Large, Voyage-4, Voyage-3-Large, Voyage-3.5, Qwen3-8B, Nemotron-8B, Gemma-3-12B and SFR-Mistral. These embeddings were used to construct row-aligned concept matrices so that each concept string, vision-aligned embedding, and language-semantic embedding referred to the same clinical observation. We use the OpenAI-text-embedding-3-large as the default text embedding for downstream analysis.

4.3 TRACE architecture for volumetric CT-language alignment

TRACE consists of a slice-wise visual encoder, an attentive volumetric pooling module and a text encoder for CT-report alignment. The model was designed to reuse strong two-dimensional visual priors while adapting them to three-dimensional CT volumes.

DINOv3-initialized slice encoder. The visual backbone was initialized from DINOv3⁶⁷, a self-supervised vision foundation model. Each axial CT slice was processed independently by the slice encoder to produce a sequence of slice-level visual embeddings. This initialization transfers general visual representations into CT without requiring a three-dimensional model to be trained from scratch.

Attentive aggregation across axial slices. To convert slice-level features into a volume-level representation, TRACE uses attentive aggregation across the depth dimension. For a CT volume with D axial slices, the slice encoder produces

$$\mathbf{H} = [h_1, \dots, h_D]^\top \in \mathbb{R}^{D \times d}, \quad (2)$$

where h_d denotes the embedding of slice d . A shallow transformer module applies self-attention along the slice index, allowing information exchange across neighbouring and distant slices⁶⁸. A learnable query vector then attends over the contextualized slice sequence to produce a fixed-length volume embedding, analogous to attention-based multiple-instance learning⁶⁹:

$$q_i = \text{AttnPool}(\mathbf{q}, \mathbf{H}_i^{\text{ctx}}) \in \mathbb{R}^d, \quad (3)$$

where $\mathbf{H}_i^{\text{ctx}}$ is the contextualized slice sequence and $\mathbf{q}$ is a learnable query. A linear projection maps q_i to the shared image-text embedding dimension, followed by ℓ_2 normalization.

Language encoder and semantic embedding pathway. The language pathway follows the large-scale unsupervised language-modelling paradigm⁷¹. During contrastive pretraining, radiology reports were tokenized and encoded into report-level text embeddings aligned with CT volume embeddings. During concept-based inference, LLM-derived concept embeddings were used to organize radiological observations, disease descriptions, and EHR-derived phenotype terms in a shared semantic space^{72,73,91,92}.

Task-agnostic CT-report contrastive pretraining TRACE was pretrained on paired CT volumes and radiology reports from CT-RATE¹⁹. Let $u_i = f_\theta(X_i)$ denote the normalized image embedding of CT volume X_i , and let $v_i = g_\phi(R_i)$ denote the normalized text embedding of its corresponding report R_i . For a minibatch of size M , we computed all pairwise image-text cosine similarities and optimized a symmetric contrastive loss⁷⁴:

$$\mathcal{L} = \frac{1}{2M} \sum_{i=1}^M \left[-\log \frac{\exp(su_i^\top v_i)}{\sum_{j=1}^M \exp(su_i^\top v_j)} - \log \frac{\exp(sv_i^\top u_i)}{\sum_{j=1}^M \exp(sv_i^\top u_j)} \right], \quad (4)$$

where $s > 0$ is a learned logit temperature. Matched CT-report pairs were treated as positives, whereas all non-matching cross-pair combinations in the minibatch served as negatives. During distributed training, embeddings were gathered across accelerators before the softmax to increase the effective number of negatives.

Model parameters were optimized using AdamW¹⁰⁸, mixed-precision arithmetic, linear learning-rate warmup, and cosine learning-rate decay. Unless otherwise stated, the slice encoder, attentive pooling module, projection layers, language encoder, and temperature parameter were optimized jointly. Checkpoints were selected using held-out CT-RATE validation performance.

Zero-shot and supervised transfer with TRACE. For TRACE-style zero-shot inference, class definitions were also represented in the LLM semantic space. For each radiological finding or disease label c , an LLM was used to generate paired affirmative and negated prompts describing the presence and absence of the target phenotype. Let p_c^+ and p_c^- denote the normalized LLM embeddings of the positive and negative prompts, respectively. For a CT volume represented by $\mathbf{z}_i^{(\text{TRACE})}$, the zero-shot class margin was computed as

$$m_{i,c} = \mathbf{z}_i^{(\text{TRACE})\top} p_c^+ - \mathbf{z}_i^{(\text{TRACE})\top} p_c^-. \quad (5)$$

These margins were used as continuous one-versus-rest class scores for AUROC computation. Thus, TRACE-style zero-shot inference first computes CT-concept similarities in the original contrastive image-text space, projects the concept profile into the LLM concept space, and then compares the projected representation with LLM-generated positive and negative disease prompts.

For supervised transfer, we trained linear probes on frozen representations. We compared direct DINOv3 visual features⁶⁷, raw CBM concept similarities $\mathbf{S}$ and LLM-projected TRACE concept features $\mathbf{Z}^{(\text{TRACE})}$. Let $\mathbf{F} \in \mathbb{R}^{N \times d}$ denote one of these frozen feature matrices after standardization on the training split. A multilabel linear probe maps $\mathbf{F}$ to class probabilities by

$$\hat{\mathbf{Y}} = \sigma(\mathbf{F}\mathbf{W}^\top + \mathbf{1}\mathbf{b}^\top), \quad \mathbf{W} \in \mathbb{R}^{C \times d}, \quad \mathbf{b} \in \mathbb{R}^C, \quad (6)$$

where C is the number of target phenotypes and $\sigma(\cdot)$ is the element-wise sigmoid function. The probe was optimized with binary cross-entropy,

$$\mathcal{L}_{\text{LP}} = -\frac{1}{NC} \sum_{i=1}^N \sum_{c=1}^C [y_{i,c} \log \hat{y}_{i,c} + (1 - y_{i,c}) \log(1 - \hat{y}_{i,c})], \quad (7)$$

using AdamW optimization¹⁰⁸. For TRACE, $\mathbf{F} = \mathbf{Z}^{(\text{TRACE})}$, so supervision is applied only after CT-concept similarities have been projected into the LLM semantic space. For each class, one-versus-rest AUROC was computed on the held-out test split, and selected experiments were repeated across multiple random seeds to estimate stochastic variation. It is worth noting that all the linear probing methods were trained with 20 different seeds for reliable importance analysis.

4.4 Concept importance analysis and disease-level audit

Projecting classifier weights back to natural-language concepts. For each trained linear model, we computed class-specific concept importance. In the original CBM, input dimensions correspond directly to concepts, so the class-concept importance matrix is the classifier weight matrix:

$$\mathbf{I}^{(\text{CBM})} = \mathbf{W}. \quad (8)$$

For TRACE, the classifier operates on LLM-projected concept features. We therefore projected classifier weights back to the original concept axes:

$$\mathbf{I}^{(\text{TRACE})} = \mathbf{W}\tilde{\mathbf{E}}_{\text{LLM}}^\top \in \mathbb{R}^{C \times K}, \quad (9)$$

where C is the number of downstream classes and K is the number of concepts. This produces an importance score for every natural-language concept and every disease label. Across random seeds, importance scores were summarized using mean absolute values and standard deviations.

Case-level concept profiling and attention-based CT visualization. For qualitative interpretation, TRACE combines ranked concept profiles with slice-level attention visualization. For each CT volume, concept similarities were computed as

$$s_{i,k} = \mathbf{u}_i^\top \tilde{\mathbf{c}}_k, \quad (10)$$

where $\mathbf{u}_i$ is the normalized image embedding for volume i and $\tilde{\mathbf{c}}_k$ is the normalized embedding of concept k . The highest-ranked concepts were displayed as a compact natural-language summary of the CT examination. To localize image evidence, we used an MST-style attention visualization strategy⁷⁹. For each axial slice, class-token-to-patch attention from the final transformer block was averaged across heads, reshaped to the patch grid and upsampled to the slice resolution. Let $\alpha_{z,h,w}$ denote the resulting attention map for slice z . Slice salience was summarized as

$$S_z = \max_{h,w} \alpha_{z,h,w}. \quad (11)$$

The most informative slices were selected by descending S_z subject to a minimum slice-index separation to avoid redundant adjacent planes. Attention overlays were used only for qualitative inspection and did not affect model prediction.

4.5 Role of human experts in concept audit, concept refinement and reader evaluation

Clinician annotation of model-discovered concepts. Human experts were first involved to determine whether the high-importance concepts discovered by TRACE reflected clinically meaningful radiology-disease associations. For each disease in CT-RATE and INSPECT, we extracted the top ten concepts ranked by linear-probe concept importance, yielding $(18 + 97) \times$

10 = 1150 disease-concept pairs. Clinicians assigned each concept to one of four predefined categories: expected evidence, defined as known and strongly supportive imaging evidence for the target disease; confounding evidence, defined as non-specific or correlative findings that may co-occur with the disease but are insufficient for diagnosis; candidate novel evidence, defined as imaging information not currently considered part of the established disease definition but potentially relevant for hypothesis generation; and artefactual evidence, defined as negated, normal, template-derived or otherwise misleading concepts. This annotation was used to assess whether TRACE recovered established radiological knowledge and whether it could surface candidate imaging findings that may represent previously under-recognized biomarkers.

Expert scoring of concept-disease relevance. Human experts were also used to validate the relevance-scoring procedure used for agentic concept refinement. We selected five diseases from CT-RATE and five diseases from INSPECT; for each disease, ten high-importance concepts were sampled for review. Two clinicians independently scored the association between each concept and its corresponding disease. These clinician scores were then compared with relevance scores generated by gpt-5.4-nano. This comparison was designed to test whether the LLM-based judging step provided a reasonable approximation of expert clinical relevance, supporting its use as a scalable component for subsequent agent-driven concept filtering and refinement.

Radiologist reader study on INSPECT CT volumes. Finally, we conducted a reader study to compare TRACE predictions with expert CT interpretation. Two radiologists, each with more than five years of experience, independently reviewed 149 CT volumes from the INSPECT test cohort and indicated whether imaging findings supported the presence of selected EHR-derived phenotypes. Their categorical readings were compared with structured labels and TRACE predictions on the same cases, without model retraining, recalibration or threshold adjustment. This evaluation tested whether TRACE could achieve performance comparable to, or better than, human experts on clinically relevant opportunistic phenotypes while preserving a transparent concept-based explanation for each prediction.

4.6 LLM-assisted concept scoring and concept-bank refinement

LLM scoring of high-importance concepts. To support a large-scale review of the concept bank, high-importance concepts were scored using an LLM-assisted (gpt-5.4-nano) judging procedure. For each disease, the model received batched prompts containing exact concept strings and returned structured JSON outputs with integer clinical relevance scores. In selected analyses, the LLM also assigned concept types corresponding to expected evidence, confounding evidence, candidate novel evidence or artefactual evidence. These LLM outputs were cached by disease and merged with concept-importance files. LLM scoring was used as a scalable audit aid and not as a substitute for clinician review.

Probe-based and importance-based concept refinement. We evaluated two strategies for constructing compact disease-relevant concept banks. In probe-based refinement, frozen linear-probe weights $\mathbf{W}_{LP} \in \mathbb{R}^{C \times d_{img}}$ were used to rank candidate concepts by

$$a_{c,k} = \mathbf{w}_c^\top \mathbf{e}_k^{\text{vis}}, \quad (12)$$

and the top concepts for each disease were retained. In importance-based refinement, full-bank CBM or TRACE models were trained across multiple seeds, class-specific importance matrices were averaged and the highest-ranked concepts were selected for each disease. Duplicate concepts selected for multiple diseases were merged while preserving the maximum score. The selected concept strings and aligned LLM embeddings were exported as refined concept banks for downstream analysis.

Counterfactual pruning of low-relevance concepts. We further tested whether audit information could be used to edit the concept representation. For a disease c , concepts with relevance scores $r_{c,k}$ below a prespecified threshold were removed, whereas concepts above the threshold were retained. Unless otherwise stated, the default threshold was $r_{c,k} \geq 3$. When only a subset of concepts had been scored, unscored concepts were retained to avoid unintended deletion. Binary linear probes were retrained on full and pruned concept banks, and performance was compared across random seeds.

4.7 Evaluation metrics

Multi-label classification performance was assessed using the area under the receiver operating characteristic curve (AUROC). AUROC was computed independently for each class and summarized as a macro-average across evaluable classes. Classes with no positive samples in a given split were excluded from macro-average computation. For threshold-dependent metrics, thresholds were selected on the validation split and then applied to the test split. When label files contained uncertain or ignored entries, those entries were excluded from metric computation.

For zero-shot evaluation, continuous class scores were obtained from paired positive-versus-negative prompt margins. For supervised linear probing, continuous probabilities were obtained from sigmoid outputs of the trained linear head. For multi-seed experiments, performance was reported as the mean and standard deviation across seeds.

4.8 Statistical analysis

Unless otherwise stated, all reported results were computed on held-out test data. For multi-seed experiments, we summarized performance as mean $\pm$ standard deviation across random seeds. For reader-study analyses and selected model comparisons, 95% confidence intervals were estimated using non-parametric bootstrap resampling over patients. Two-sided 500 times bootstrap P values were computed for differences in AUROC or operating-point performance.

4.9 Computing hardware and software

All experiments were conducted on NVIDIA GH200 systems. Analyses were implemented in Python 3.10.16 using PyTorch 2.10.0 with CUDA 13.0 support for model training and inference. NumPy 2.2.6 and Pandas 2.3.3 were used for data processing, Scikit-learn 1.7.2 and SciPy 1.15.3 for classical machine-learning utilities and metric computation, and Matplotlib 3.10.6 for visualization. The modeling stack also used Torchvision 0.25.0, Hugging Face Transformers 4.57.1, Sentence-Transformers 5.1.2, and OpenCLIP 3.3.0.

5 Data availability

The datasets used in this study are publicly available from their respective repositories. CT-RATE¹⁹ is available at <https://huggingface.co/datasets/ibrahimhamamci/CT-RATE>. INSPECT⁷⁰ is available at <https://inspect.stanford.edu/>. Merlin² is available at <https://stanfordaimi.azurewebsites.net/datasets/60b9c7ff-877b-48ce-96c3-0194c8205c>. RAD-ChestCT¹⁰ is available at <https://zenodo.org/records/6406114>. Processed label tables, concept-bank files, and evaluation outputs used to reproduce the main results will be released with the study.

6 Code availability

The source code for training the CLIP is available at https://github.com/peterhan91/clip_3d_ct. The source code will be made available at <https://github.com/haifangong/TRACE> after major revision. Trained checkpoints, concept banks, and evaluation results will be made available at <https://huggingface.co/datasets/haifan-gong/TRACE>.

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None.

8 Author contributions

T.H., H.G., R.W., C.D., L.S. and D.T. conceived the study. T.H. and H.G. designed and implemented the TRACE architecture. H.G. performed the concept extraction and embedding experiments, zero-shot evaluation experiments, and, together with T.H., implemented and evaluated the concept bottleneck models. T.H. and R.W. conducted the visual-language pretraining, with technical support from Y.T., D.T., E.M.B., L.C.A. and K.K.B. provided clinical insights. Y.W., H.L. and K.L. performed the reader study, concept annotation, and evaluated model interpretability. W.W. and D.A.M. contributed imaging and translational research expertise. D.T., C.D., L.S. and T.H. supervised the study. H.G. and T.H. wrote the manuscript with input from all authors.

9 Competing interests

D.T. holds shares in StratifAI GmbH and reports speaker fees from Bayer, Germany. K.K.B. reports speaker fees from Canon Medical Systems Corporation and GE HealthCare. JNK declares consulting services for Owkin, France; DoMore Diagnostics, Norway; Panakeia, UK, and Scailyte, Basel, Switzerland; furthermore JNK holds shares in Kather Consulting, Dresden, Germany; and StratifAI GmbH, Dresden, Germany, and has received honoraria for lectures and advisory board participation by AstraZeneca, Bayer, Eisai, MSD, BMS, Roche, Pfizer and Fresenius. No other disclosures are reported.

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Supplementary Information

Table S1. Zero-shot phenotype prediction on INSPECT, part I. Class-wise AUROC values with 95% confidence intervals are reported for the first 49 INSPECT phenotypes. **Bold** indicates the best result in each row.

Finding	N	TRACE	Merlin	CT-CLIP	MedSigLIP	BiomedCLIP	OpenAI CLIP
Heart valve dis.	95	0.680 (0.620, 0.740)	0.542 (0.481, 0.602)	0.523 (0.463, 0.582)	0.584 (0.523, 0.645)	0.557 (0.497, 0.618)	0.488 (0.430, 0.547)
Hypotension	229	0.673 (0.634, 0.713)	0.578 (0.538, 0.618)	0.531 (0.491, 0.570)	0.571 (0.531, 0.611)	0.597 (0.557, 0.637)	0.547 (0.507, 0.586)
Shock	86	0.664 (0.601, 0.728)	0.628 (0.564, 0.692)	0.593 (0.529, 0.657)	0.473 (0.412, 0.533)	0.428 (0.369, 0.486)	0.592 (0.528, 0.656)
Hypertension	847	0.617 (0.594, 0.640)	0.411 (0.390, 0.433)	0.539 (0.516, 0.562)	0.538 (0.515, 0.561)	0.458 (0.436, 0.481)	0.510 (0.488, 0.533)
Cardiomegaly	228	0.610 (0.570, 0.650)	0.574 (0.534, 0.614)	0.546 (0.507, 0.586)	0.527 (0.487, 0.566)	0.594 (0.554, 0.634)	0.547 (0.508, 0.587)
Cardiac conduction dis.	981	0.604 (0.583, 0.626)	0.547 (0.525, 0.569)	0.566 (0.545, 0.588)	0.524 (0.502, 0.546)	0.546 (0.524, 0.568)	0.577 (0.556, 0.599)
Rheumatic valve dis.	62	0.603 (0.528, 0.679)	0.567 (0.493, 0.642)	0.487 (0.415, 0.559)	0.511 (0.437, 0.584)	0.448 (0.378, 0.518)	0.439 (0.370, 0.508)
Pulmonary heart dis.	459	0.571 (0.542, 0.600)	0.499 (0.470, 0.527)	0.545 (0.516, 0.574)	0.509 (0.480, 0.538)	0.555 (0.526, 0.584)	0.523 (0.495, 0.552)
Cardiac/circulatory anomalies	56	0.555 (0.477, 0.633)	0.492 (0.417, 0.568)	0.566 (0.488, 0.645)	0.537 (0.459, 0.615)	0.612 (0.533, 0.691)	0.563 (0.484, 0.642)
Other pulm. inflam./edema	88	0.783 (0.726, 0.841)	0.517 (0.455, 0.579)	0.600 (0.536, 0.663)	0.580 (0.517, 0.643)	0.631 (0.567, 0.694)	0.499 (0.437, 0.560)
Pneumonia	342	0.777 (0.748, 0.807)	0.635 (0.602, 0.669)	0.507 (0.474, 0.539)	0.629 (0.596, 0.663)	0.613 (0.580, 0.647)	0.501 (0.469, 0.534)
Empyema/pneumothorax	92	0.757 (0.699, 0.815)	0.668 (0.606, 0.729)	0.500 (0.440, 0.559)	0.620 (0.558, 0.682)	0.706 (0.646, 0.767)	0.516 (0.455, 0.576)
Aspiration pneumonitis	123	0.743 (0.692, 0.794)	0.700 (0.648, 0.753)	0.533 (0.480, 0.586)	0.607 (0.553, 0.661)	0.601 (0.547, 0.654)	0.621 (0.567, 0.675)
Pulm. congestion/hypostasis	207	0.740 (0.700, 0.779)	0.589 (0.547, 0.631)	0.558 (0.516, 0.600)	0.573 (0.531, 0.614)	0.681 (0.640, 0.722)	0.574 (0.533, 0.616)
Other respiratory dis.	79	0.716 (0.651, 0.780)	0.483 (0.419, 0.547)	0.544 (0.478, 0.610)	0.535 (0.469, 0.601)	0.572 (0.505, 0.638)	0.491 (0.427, 0.555)
Asphyxia/hypoxemia	357	0.706 (0.675, 0.738)	0.645 (0.613, 0.678)	0.563 (0.530, 0.595)	0.551 (0.518, 0.583)	0.601 (0.568, 0.634)	0.559 (0.526, 0.591)
Pleurisy/pleural effusion	576	0.703 (0.678, 0.729)	0.554 (0.527, 0.580)	0.564 (0.537, 0.590)	0.572 (0.545, 0.599)	0.675 (0.649, 0.701)	0.523 (0.497, 0.549)
Chronic airway obstruction	261	0.683 (0.646, 0.720)	0.611 (0.574, 0.649)	0.529 (0.492, 0.566)	0.546 (0.509, 0.584)	0.532 (0.495, 0.570)	0.567 (0.530, 0.605)
Abnormal lung findings	957	0.656 (0.635, 0.678)	0.492 (0.470, 0.514)	0.537 (0.515, 0.560)	0.536 (0.514, 0.558)	0.572 (0.550, 0.594)	0.509 (0.487, 0.531)
Pulm. collapse/emphysema	342	0.635 (0.602, 0.668)	0.482 (0.450, 0.514)	0.532 (0.499, 0.565)	0.528 (0.495, 0.561)	0.622 (0.589, 0.656)	0.526 (0.494, 0.559)
Other lung dis.	144	0.630 (0.580, 0.680)	0.554 (0.505, 0.604)	0.473 (0.425, 0.520)	0.550 (0.501, 0.600)	0.558 (0.509, 0.608)	0.502 (0.454, 0.551)
Abnormal sputum	58	0.616 (0.538, 0.694)	0.626 (0.548, 0.703)	0.523 (0.447, 0.599)	0.534 (0.458, 0.611)	0.654 (0.577, 0.732)	0.609 (0.531, 0.687)
Respiratory abnormalities	93	0.607 (0.545, 0.668)	0.524 (0.463, 0.584)	0.549 (0.488, 0.610)	0.487 (0.428, 0.546)	0.592 (0.531, 0.654)	0.509 (0.449, 0.569)
Other upper resp. dis.	50	0.601 (0.517, 0.685)	0.564 (0.481, 0.647)	0.537 (0.454, 0.619)	0.596 (0.513, 0.680)	0.558 (0.475, 0.641)	0.537 (0.454, 0.619)
Asthma	189	0.542 (0.499, 0.585)	0.493 (0.451, 0.535)	0.508 (0.465, 0.550)	0.502 (0.459, 0.544)	0.499 (0.456, 0.541)	0.517 (0.474, 0.560)
Venous embolism/thrombosis	235	0.622 (0.583, 0.662)	0.540 (0.501, 0.579)	0.588 (0.549, 0.628)	0.568 (0.528, 0.607)	0.558 (0.519, 0.597)	0.518 (0.479, 0.556)
Cerebrovascular dis.	108	0.595 (0.538, 0.652)	0.526 (0.470, 0.582)	0.555 (0.498, 0.612)	0.626 (0.569, 0.684)	0.488 (0.433, 0.543)	0.526 (0.470, 0.582)
Other aneurysm	55	0.568 (0.488, 0.647)	0.532 (0.453, 0.610)	0.571 (0.491, 0.650)	0.472 (0.397, 0.548)	0.481 (0.405, 0.557)	0.595 (0.515, 0.675)
Prostate hyperplasia	85	0.652 (0.588, 0.716)	0.550 (0.486, 0.613)	0.566 (0.502, 0.630)	0.528 (0.465, 0.591)	0.532 (0.469, 0.596)	0.475 (0.413, 0.536)
Other urinary disorders	156	0.625 (0.577, 0.673)	0.413 (0.370, 0.457)	0.568 (0.521, 0.616)	0.555 (0.508, 0.603)	0.588 (0.540, 0.636)	0.540 (0.492, 0.587)
Kidney/ureter disorders	69	0.571 (0.500, 0.642)	0.547 (0.477, 0.618)	0.530 (0.460, 0.600)	0.501 (0.432, 0.570)	0.582 (0.511, 0.653)	0.516 (0.447, 0.586)
Protein-calorie malnutrition	271	0.682 (0.646, 0.719)	0.475 (0.440, 0.511)	0.562 (0.525, 0.599)	0.637 (0.600, 0.673)	0.532 (0.495, 0.568)	0.556 (0.519, 0.593)
Diabetes mellitus	434	0.657 (0.628, 0.687)	0.446 (0.418, 0.475)	0.548 (0.518, 0.578)	0.541 (0.511, 0.571)	0.491 (0.462, 0.520)	0.494 (0.464, 0.523)
Mineral metabolism dis.	139	0.631 (0.581, 0.682)	0.544 (0.494, 0.594)	0.525 (0.476, 0.575)	0.537 (0.487, 0.587)	0.542 (0.492, 0.592)	0.526 (0.476, 0.575)
Obesity/overnutrition	224	0.550 (0.510, 0.590)	0.356 (0.322, 0.390)	0.561 (0.521, 0.601)	0.385 (0.350, 0.421)	0.442 (0.404, 0.479)	0.444 (0.407, 0.482)
Ascites	122	0.743 (0.692, 0.794)	0.705 (0.653, 0.758)	0.553 (0.499, 0.607)	0.557 (0.504, 0.611)	0.606 (0.552, 0.660)	0.513 (0.460, 0.565)
Other peritoneal dis.	59	0.741 (0.668, 0.814)	0.649 (0.572, 0.725)	0.658 (0.581, 0.735)	0.520 (0.445, 0.596)	0.620 (0.543, 0.697)	0.613 (0.535, 0.690)
Dysphagia	138	0.699 (0.650, 0.749)	0.587 (0.536, 0.638)	0.559 (0.508, 0.610)	0.642 (0.591, 0.692)	0.583 (0.532, 0.633)	0.542 (0.492, 0.593)
Digestive symptoms	193	0.684 (0.641, 0.726)	0.556 (0.513, 0.600)	0.569 (0.526, 0.613)	0.522 (0.479, 0.564)	0.559 (0.516, 0.602)	0.477 (0.436, 0.519)
Pancreatic diseases	58	0.680 (0.603, 0.757)	0.520 (0.445, 0.596)	0.500 (0.425, 0.575)	0.542 (0.466, 0.619)	0.521 (0.445, 0.597)	0.478 (0.405, 0.552)
Other intestinal dis.	111	0.671 (0.615, 0.727)	0.609 (0.552, 0.666)	0.637 (0.580, 0.693)	0.493 (0.438, 0.547)	0.610 (0.553, 0.667)	0.625 (0.568, 0.681)
Abdominal hernia	88	0.668 (0.605, 0.731)	0.548 (0.485, 0.611)	0.564 (0.501, 0.627)	0.555 (0.493, 0.618)	0.542 (0.480, 0.605)	0.523 (0.461, 0.585)
Constipation	201	0.656 (0.613, 0.698)	0.513 (0.472, 0.555)	0.540 (0.498, 0.582)	0.526 (0.485, 0.568)	0.453 (0.413, 0.493)	0.499 (0.458, 0.540)
Esophageal diseases	342	0.635 (0.602, 0.668)	0.525 (0.492, 0.558)	0.534 (0.501, 0.567)	0.565 (0.532, 0.598)	0.509 (0.477, 0.542)	0.518 (0.485, 0.551)
Functional digestive dis.	81	0.626 (0.560, 0.692)	0.560 (0.494, 0.625)	0.526 (0.461, 0.590)	0.509 (0.445, 0.573)	0.546 (0.481, 0.611)	0.524 (0.460, 0.589)
Other liver dis.	212	0.614 (0.572, 0.655)	0.545 (0.504, 0.586)	0.544 (0.503, 0.585)	0.471 (0.432, 0.511)	0.576 (0.534, 0.617)	0.524 (0.483, 0.564)
Nausea/vomiting	188	0.578 (0.534, 0.622)	0.557 (0.513, 0.600)	0.534 (0.491, 0.577)	0.570 (0.526, 0.613)	0.537 (0.494, 0.580)	0.533 (0.489, 0.576)
Abdominal pain	274	0.571 (0.535, 0.608)	0.521 (0.485, 0.558)	0.551 (0.514, 0.588)	0.476 (0.441, 0.511)	0.534 (0.497, 0.570)	0.473 (0.438, 0.508)
White blood cell dis.	194	0.671 (0.629, 0.714)	0.519 (0.476, 0.561)	0.549 (0.506, 0.591)	0.506 (0.464, 0.548)	0.526 (0.483, 0.568)	0.506 (0.464, 0.548)

Table S2. Zero-shot phenotype prediction on INSPECT, part II. Class-wise AUROC values with 95% confidence intervals are reported for the remaining 48 INSPECT phenotypes. **Bold** indicates the best result in each row.

Finding	N	TRACE	Merlin	CT-CLIP	MedSigLIP	BiomedCLIP	OpenAI CLIP
Other anemias	473	0.659 (0.631, 0.688)	0.659 (0.630, 0.687)	0.557 (0.528, 0.586)	0.512 (0.484, 0.541)	0.558 (0.529, 0.587)	0.528 (0.500, 0.557)
Purpura/hemorrhagic dis.	134	0.655 (0.604, 0.706)	0.573 (0.522, 0.625)	0.609 (0.558, 0.661)	0.557 (0.506, 0.609)	0.538 (0.487, 0.589)	0.535 (0.484, 0.586)
Aplastic anemia	78	0.608 (0.541, 0.675)	0.618 (0.551, 0.686)	0.554 (0.488, 0.621)	0.472 (0.409, 0.536)	0.496 (0.431, 0.561)	0.506 (0.441, 0.572)
Other hematologic dis.	171	0.569 (0.523, 0.615)	0.565 (0.519, 0.611)	0.525 (0.480, 0.570)	0.515 (0.471, 0.560)	0.545 (0.499, 0.590)	0.554 (0.509, 0.600)
Nonspecific blood findings	186	0.563 (0.519, 0.607)	0.471 (0.429, 0.513)	0.512 (0.469, 0.555)	0.513 (0.470, 0.556)	0.513 (0.470, 0.556)	0.491 (0.448, 0.533)
Cellulitis/abscess	55	0.636 (0.556, 0.716)	0.505 (0.427, 0.582)	0.580 (0.501, 0.660)	0.559 (0.479, 0.638)	0.560 (0.481, 0.639)	0.557 (0.477, 0.636)
Urinary tract infection	90	0.609 (0.546, 0.671)	0.550 (0.488, 0.612)	0.582 (0.520, 0.645)	0.541 (0.479, 0.603)	0.500 (0.440, 0.561)	0.540 (0.478, 0.602)
Respiratory cancer	145	0.763 (0.717, 0.809)	0.630 (0.581, 0.680)	0.482 (0.435, 0.530)	0.518 (0.469, 0.566)	0.547 (0.498, 0.596)	0.500 (0.452, 0.548)
Suspected/other cancer	179	0.613 (0.568, 0.658)	0.533 (0.489, 0.577)	0.551 (0.506, 0.595)	0.495 (0.452, 0.539)	0.607 (0.562, 0.652)	0.527 (0.483, 0.571)
Leukemia	62	0.560 (0.485, 0.635)	0.517 (0.443, 0.590)	0.474 (0.403, 0.546)	0.513 (0.440, 0.587)	0.547 (0.472, 0.621)	0.494 (0.422, 0.566)
Secondary malignancy	372	0.542 (0.510, 0.574)	0.611 (0.579, 0.643)	0.486 (0.455, 0.517)	0.549 (0.517, 0.581)	0.579 (0.547, 0.611)	0.545 (0.513, 0.577)
Breast cancer	90	0.524 (0.463, 0.586)	0.607 (0.545, 0.670)	0.461 (0.402, 0.520)	0.491 (0.431, 0.552)	0.529 (0.467, 0.590)	0.524 (0.463, 0.585)
Neurological dis.	200	0.661 (0.619, 0.703)	0.508 (0.466, 0.549)	0.588 (0.545, 0.630)	0.540 (0.497, 0.582)	0.577 (0.534, 0.619)	0.595 (0.552, 0.637)
Dementia/cognitive dis.	158	0.644 (0.596, 0.691)	0.420 (0.377, 0.463)	0.548 (0.500, 0.595)	0.599 (0.552, 0.647)	0.549 (0.502, 0.596)	0.647 (0.599, 0.695)
Inflammatory/toxic neuropathy	51	0.641 (0.559, 0.724)	0.565 (0.483, 0.647)	0.483 (0.404, 0.562)	0.492 (0.412, 0.571)	0.508 (0.428, 0.588)	0.487 (0.408, 0.566)
Other brain conditions	140	0.639 (0.589, 0.690)	0.569 (0.518, 0.619)	0.574 (0.524, 0.625)	0.554 (0.504, 0.604)	0.622 (0.571, 0.673)	0.520 (0.470, 0.569)
Sleep dis.	295	0.638 (0.602, 0.673)	0.486 (0.451, 0.520)	0.569 (0.533, 0.604)	0.555 (0.519, 0.590)	0.493 (0.459, 0.528)	0.486 (0.451, 0.520)
Anxiety dis.	270	0.630 (0.593, 0.667)	0.518 (0.481, 0.554)	0.523 (0.486, 0.559)	0.513 (0.477, 0.549)	0.531 (0.495, 0.568)	0.513 (0.477, 0.549)
Adjustment reaction	64	0.626 (0.552, 0.700)	0.449 (0.380, 0.518)	0.546 (0.473, 0.619)	0.554 (0.481, 0.628)	0.424 (0.357, 0.491)	0.495 (0.423, 0.566)
Epilepsy/seizures	74	0.595 (0.526, 0.664)	0.484 (0.418, 0.550)	0.522 (0.455, 0.590)	0.579 (0.510, 0.647)	0.521 (0.454, 0.589)	0.537 (0.469, 0.605)
Mood dis.	242	0.590 (0.551, 0.629)	0.475 (0.438, 0.513)	0.577 (0.538, 0.616)	0.546 (0.508, 0.585)	0.495 (0.457, 0.533)	0.542 (0.504, 0.581)
Substance addiction/dis.	81	0.561 (0.496, 0.627)	0.534 (0.469, 0.598)	0.596 (0.530, 0.662)	0.467 (0.405, 0.529)	0.528 (0.463, 0.593)	0.512 (0.448, 0.576)
Tobacco use dis.	125	0.555 (0.502, 0.608)	0.518 (0.466, 0.570)	0.524 (0.471, 0.576)	0.499 (0.447, 0.551)	0.502 (0.450, 0.554)	0.557 (0.504, 0.610)
Alcohol-related dis.	86	0.551 (0.487, 0.614)	0.615 (0.551, 0.680)	0.602 (0.538, 0.666)	0.455 (0.395, 0.515)	0.578 (0.514, 0.642)	0.551 (0.487, 0.614)
Other headache syndromes	50	0.475 (0.395, 0.554)	0.539 (0.457, 0.622)	0.554 (0.471, 0.637)	0.567 (0.484, 0.651)	0.518 (0.436, 0.599)	0.517 (0.435, 0.598)
Vestibular dis.	94	0.425 (0.370, 0.481)	0.436 (0.380, 0.492)	0.498 (0.439, 0.557)	0.529 (0.468, 0.589)	0.485 (0.426, 0.544)	0.478 (0.420, 0.536)
Transplant complications	126	0.634 (0.581, 0.687)	0.470 (0.420, 0.520)	0.659 (0.606, 0.712)	0.590 (0.537, 0.643)	0.652 (0.600, 0.705)	0.621 (0.568, 0.675)
Muscle symptoms	74	0.624 (0.555, 0.693)	0.551 (0.483, 0.619)	0.607 (0.538, 0.676)	0.582 (0.513, 0.650)	0.618 (0.549, 0.687)	0.514 (0.447, 0.581)
MSK nonspecific findings	53	0.609 (0.527, 0.690)	0.427 (0.353, 0.501)	0.482 (0.405, 0.560)	0.597 (0.515, 0.678)	0.520 (0.440, 0.599)	0.514 (0.435, 0.593)
Osteoarthritis	85	0.603 (0.538, 0.667)	0.597 (0.533, 0.662)	0.571 (0.507, 0.635)	0.569 (0.505, 0.633)	0.464 (0.404, 0.525)	0.530 (0.467, 0.593)
Back pain	124	0.560 (0.507, 0.614)	0.478 (0.427, 0.529)	0.504 (0.452, 0.556)	0.535 (0.483, 0.588)	0.497 (0.445, 0.548)	0.477 (0.426, 0.528)
Joint pain	108	0.537 (0.480, 0.593)	0.505 (0.450, 0.561)	0.512 (0.457, 0.568)	0.493 (0.438, 0.548)	0.528 (0.472, 0.585)	0.509 (0.453, 0.565)
Injury	58	0.536 (0.459, 0.612)	0.481 (0.407, 0.555)	0.521 (0.445, 0.597)	0.548 (0.471, 0.625)	0.522 (0.446, 0.598)	0.493 (0.418, 0.567)
Limb pain	99	0.516 (0.457, 0.574)	0.483 (0.426, 0.540)	0.526 (0.468, 0.585)	0.575 (0.515, 0.635)	0.548 (0.489, 0.607)	0.489 (0.431, 0.546)
Limb MSK symptoms	121	0.506 (0.453, 0.558)	0.443 (0.393, 0.493)	0.550 (0.496, 0.604)	0.513 (0.460, 0.565)	0.521 (0.468, 0.574)	0.502 (0.450, 0.555)
Other soft tissue dis.	108	0.490 (0.435, 0.545)	0.525 (0.468, 0.581)	0.559 (0.502, 0.616)	0.531 (0.475, 0.587)	0.550 (0.493, 0.607)	0.533 (0.476, 0.589)
Immune disorders	68	0.715 (0.645, 0.784)	0.570 (0.498, 0.641)	0.590 (0.518, 0.662)	0.621 (0.549, 0.693)	0.526 (0.456, 0.597)	0.610 (0.538, 0.682)
Procedure complications	58	0.687 (0.611, 0.764)	0.600 (0.522, 0.678)	0.648 (0.570, 0.726)	0.640 (0.563, 0.718)	0.657 (0.580, 0.734)	0.660 (0.583, 0.738)
Ill-defined morbidity	60	0.659 (0.583, 0.735)	0.555 (0.479, 0.630)	0.552 (0.476, 0.628)	0.503 (0.429, 0.577)	0.559 (0.483, 0.635)	0.580 (0.504, 0.657)
Malaise/fatigue	307	0.628 (0.593, 0.662)	0.569 (0.534, 0.604)	0.559 (0.524, 0.593)	0.518 (0.484, 0.553)	0.583 (0.548, 0.618)	0.540 (0.506, 0.575)
Fever unknown origin	209	0.614 (0.572, 0.656)	0.534 (0.493, 0.575)	0.533 (0.492, 0.574)	0.527 (0.486, 0.568)	0.546 (0.505, 0.587)	0.465 (0.425, 0.504)
Acquired organ absence	163	0.614 (0.567, 0.661)	0.585 (0.538, 0.632)	0.547 (0.501, 0.594)	0.539 (0.492, 0.585)	0.530 (0.484, 0.577)	0.467 (0.423, 0.512)
Toxic solvent/alcohol effect	135	0.608 (0.557, 0.660)	0.525 (0.474, 0.575)	0.546 (0.495, 0.596)	0.514 (0.464, 0.564)	0.565 (0.514, 0.617)	0.534 (0.483, 0.584)
Skin symptoms	90	0.560 (0.498, 0.622)	0.441 (0.383, 0.499)	0.523 (0.461, 0.584)	0.525 (0.464, 0.587)	0.548 (0.486, 0.610)	0.508 (0.447, 0.569)
Abdomen/pelvis symptoms	65	0.554 (0.481, 0.627)	0.571 (0.498, 0.644)	0.588 (0.514, 0.661)	0.470 (0.401, 0.539)	0.530 (0.458, 0.603)	0.525 (0.453, 0.597)
Other tests	97	0.553 (0.493, 0.613)	0.550 (0.491, 0.610)	0.556 (0.496, 0.616)	0.495 (0.437, 0.553)	0.557 (0.497, 0.617)	0.526 (0.467, 0.585)
Syncope/collapse	98	0.459 (0.403, 0.515)	0.534 (0.474, 0.593)	0.519 (0.460, 0.578)	0.516 (0.458, 0.575)	0.515 (0.456, 0.573)	0.561 (0.502, 0.621)
Nonspecific chest pain	701	0.448 (0.424, 0.472)	0.527 (0.503, 0.552)	0.496 (0.472, 0.520)	0.509 (0.485, 0.534)	0.490 (0.466, 0.515)	0.473 (0.449, 0.497)

Table S3. Zero-shot finding prediction on CT-RATE. Class-wise AUROC values with 95% confidence intervals are reported for the 18 CT-RATE findings. **Bold** indicates the best result in each row.

Finding	N	TRACE	Merlin	CT-CLIP	MedSigLIP	BiomedCLIP	OpenAI CLIP
Cardiomegaly	273	0.824 (0.793, 0.855)	0.673 (0.637, 0.710)	0.851 (0.822, 0.880)	0.799 (0.767, 0.832)	0.519 (0.482, 0.555)	0.573 (0.536, 0.610)
Pericardial effusion	205	0.779 (0.740, 0.817)	0.602 (0.559, 0.644)	0.779 (0.740, 0.817)	0.638 (0.596, 0.681)	0.596 (0.554, 0.639)	0.519 (0.477, 0.561)
Arterial wall calcification	733	0.856 (0.838, 0.874)	0.640 (0.615, 0.664)	0.841 (0.822, 0.860)	0.727 (0.704, 0.750)	0.555 (0.530, 0.580)	0.472 (0.448, 0.497)
Coronary artery calc.	643	0.845 (0.825, 0.864)	0.625 (0.599, 0.651)	0.846 (0.827, 0.866)	0.751 (0.727, 0.774)	0.536 (0.510, 0.562)	0.507 (0.481, 0.533)
Pleural effusion	336	0.921 (0.901, 0.941)	0.744 (0.713, 0.776)	0.901 (0.879, 0.923)	0.641 (0.607, 0.675)	0.543 (0.509, 0.577)	0.464 (0.432, 0.497)
Peribronchial thick.	312	0.712 (0.678, 0.745)	0.562 (0.527, 0.597)	0.681 (0.647, 0.715)	0.613 (0.578, 0.648)	0.551 (0.516, 0.586)	0.454 (0.421, 0.487)
Bronchiectasis	267	0.629 (0.592, 0.667)	0.560 (0.523, 0.598)	0.637 (0.600, 0.674)	0.557 (0.520, 0.595)	0.606 (0.568, 0.643)	0.472 (0.436, 0.508)
Emphysema	496	0.653 (0.625, 0.682)	0.556 (0.527, 0.585)	0.727 (0.700, 0.754)	0.588 (0.559, 0.617)	0.513 (0.485, 0.542)	0.508 (0.480, 0.537)
Atelectasis	600	0.646 (0.619, 0.672)	0.568 (0.541, 0.595)	0.683 (0.658, 0.709)	0.659 (0.632, 0.685)	0.500 (0.473, 0.527)	0.501 (0.474, 0.527)
Consolidation	490	0.824 (0.800, 0.847)	0.610 (0.581, 0.639)	0.725 (0.698, 0.752)	0.576 (0.547, 0.605)	0.531 (0.502, 0.560)	0.441 (0.413, 0.469)
Lung opacity	971	0.774 (0.754, 0.793)	0.574 (0.551, 0.597)	0.627 (0.604, 0.650)	0.572 (0.549, 0.595)	0.489 (0.465, 0.512)	0.476 (0.453, 0.500)
Lung nodule	1123	0.562 (0.540, 0.585)	0.523 (0.501, 0.546)	0.584 (0.561, 0.606)	0.544 (0.521, 0.567)	0.512 (0.489, 0.535)	0.494 (0.471, 0.516)
Pulm. fibrotic sequela	673	0.608 (0.582, 0.633)	0.558 (0.532, 0.583)	0.565 (0.539, 0.591)	0.544 (0.518, 0.570)	0.550 (0.524, 0.575)	0.537 (0.512, 0.563)
Mosaic attenuation	213	0.654 (0.613, 0.695)	0.571 (0.529, 0.613)	0.755 (0.716, 0.793)	0.590 (0.548, 0.631)	0.349 (0.314, 0.384)	0.430 (0.391, 0.468)
Interlobular septal thick.	206	0.818 (0.783, 0.854)	0.615 (0.573, 0.658)	0.774 (0.735, 0.812)	0.593 (0.551, 0.636)	0.563 (0.521, 0.605)	0.487 (0.446, 0.528)
Medical material	277	0.682 (0.646, 0.718)	0.554 (0.517, 0.591)	0.704 (0.668, 0.740)	0.568 (0.531, 0.605)	0.510 (0.474, 0.546)	0.516 (0.480, 0.553)
Lymphadenopathy	693	0.707 (0.683, 0.731)	0.525 (0.500, 0.551)	0.704 (0.680, 0.728)	0.618 (0.593, 0.643)	0.508 (0.483, 0.534)	0.476 (0.451, 0.502)
Hiatal hernia	341	0.666 (0.633, 0.699)	0.577 (0.543, 0.611)	0.706 (0.674, 0.738)	0.574 (0.540, 0.607)	0.479 (0.447, 0.512)	0.525 (0.491, 0.558)

Table S4. Zero-shot phenotype prediction on Merlin. Class-wise AUROC values with 95% confidence intervals are reported for Merlin chest and opportunistic labels. **Bold** indicates the best result in each row.

Finding	N	TRACE	Merlin	CT-CLIP	MedSigLIP	BiomedCLIP	OpenAI CLIP
Aortic valve calcification	72	0.663 (0.574, 0.752)	0.746 (0.665, 0.827)	0.627 (0.535, 0.719)	0.835 (0.768, 0.902)	0.733 (0.651, 0.816)	0.557 (0.462, 0.652)
Coronary calcification	63	0.733 (0.659, 0.807)	0.664 (0.586, 0.741)	0.672 (0.595, 0.750)	0.742 (0.669, 0.816)	0.650 (0.572, 0.728)	0.595 (0.517, 0.674)
Cardiomegaly	119	0.698 (0.639, 0.757)	0.668 (0.608, 0.728)	0.622 (0.560, 0.683)	0.728 (0.671, 0.785)	0.617 (0.555, 0.678)	0.644 (0.583, 0.705)
Atherosclerosis	879	0.689 (0.626, 0.752)	0.651 (0.584, 0.718)	0.590 (0.517, 0.662)	0.604 (0.532, 0.676)	0.594 (0.522, 0.667)	0.528 (0.451, 0.606)
Thrombosis	77	0.690 (0.587, 0.793)	0.709 (0.609, 0.809)	0.454 (0.334, 0.575)	0.510 (0.391, 0.629)	0.538 (0.420, 0.656)	0.587 (0.473, 0.702)
Abdominal aortic aneurysm	17	0.735 (0.591, 0.880)	0.742 (0.599, 0.885)	0.565 (0.411, 0.720)	0.638 (0.485, 0.792)	0.514 (0.361, 0.667)	0.481 (0.330, 0.631)
Pleural effusion	834	0.834 (0.814, 0.855)	0.832 (0.811, 0.853)	0.580 (0.550, 0.610)	0.721 (0.695, 0.748)	0.610 (0.580, 0.639)	0.544 (0.514, 0.575)
Atelectasis	903	0.530 (0.504, 0.556)	0.558 (0.532, 0.584)	0.579 (0.553, 0.605)	0.690 (0.666, 0.713)	0.489 (0.463, 0.515)	0.493 (0.467, 0.519)
Submucosal edema	144	0.592 (0.528, 0.656)	0.622 (0.559, 0.685)	0.488 (0.423, 0.553)	0.547 (0.483, 0.612)	0.617 (0.554, 0.680)	0.489 (0.424, 0.553)
Free air	45	0.664 (0.575, 0.752)	0.630 (0.540, 0.719)	0.568 (0.479, 0.657)	0.514 (0.427, 0.601)	0.502 (0.415, 0.588)	0.452 (0.368, 0.535)
Ascites	365	0.786 (0.748, 0.823)	0.831 (0.798, 0.864)	0.588 (0.539, 0.637)	0.687 (0.642, 0.732)	0.682 (0.636, 0.727)	0.512 (0.461, 0.563)
Hiatal hernia	312	0.598 (0.563, 0.634)	0.602 (0.575, 0.637)	0.611 (0.575, 0.646)	0.620 (0.584, 0.655)	0.464 (0.430, 0.498)	0.567 (0.531, 0.602)
Hepatomegaly	57	0.578 (0.498, 0.657)	0.662 (0.583, 0.741)	0.594 (0.514, 0.673)	0.641 (0.562, 0.721)	0.510 (0.433, 0.588)	0.605 (0.526, 0.685)
Splenomegaly	141	0.612 (0.562, 0.663)	0.649 (0.599, 0.699)	0.623 (0.573, 0.674)	0.587 (0.537, 0.638)	0.603 (0.553, 0.653)	0.621 (0.571, 0.671)
Pancreatic atrophy	161	0.705 (0.659, 0.751)	0.653 (0.606, 0.700)	0.579 (0.532, 0.626)	0.691 (0.644, 0.737)	0.575 (0.528, 0.622)	0.547 (0.500, 0.593)
Hepatic steatosis	272	0.471 (0.355, 0.587)	0.586 (0.479, 0.693)	0.591 (0.485, 0.697)	0.437 (0.319, 0.554)	0.417 (0.299, 0.534)	0.437 (0.320, 0.554)
Gallstones	148	0.574 (0.516, 0.633)	0.546 (0.487, 0.605)	0.523 (0.464, 0.582)	0.634 (0.576, 0.691)	0.556 (0.497, 0.615)	0.504 (0.445, 0.563)
Surgically absent gallbladder	843	0.429 (0.407, 0.452)	0.579 (0.556, 0.602)	0.552 (0.529, 0.575)	0.608 (0.585, 0.631)	0.594 (0.571, 0.617)	0.538 (0.515, 0.561)
Appendicitis	43	0.440 (0.324, 0.556)	0.606 (0.491, 0.721)	0.581 (0.465, 0.696)	0.515 (0.398, 0.632)	0.538 (0.421, 0.656)	0.461 (0.344, 0.577)
Bowel obstruction	72	0.680 (0.610, 0.750)	0.587 (0.516, 0.657)	0.522 (0.453, 0.591)	0.540 (0.470, 0.609)	0.489 (0.421, 0.557)	0.534 (0.465, 0.604)
Biliary ductal dilation	52	0.611 (0.523, 0.700)	0.559 (0.470, 0.648)	0.438 (0.353, 0.523)	0.532 (0.444, 0.621)	0.407 (0.325, 0.490)	0.524 (0.435, 0.612)
Osteopenia	147	0.605 (0.554, 0.656)	0.745 (0.698, 0.793)	0.491 (0.441, 0.540)	0.868 (0.830, 0.906)	0.335 (0.293, 0.377)	0.374 (0.330, 0.419)
Fracture	177	0.618 (0.561, 0.675)	0.591 (0.533, 0.648)	0.528 (0.469, 0.586)	0.733 (0.682, 0.784)	0.616 (0.558, 0.673)	0.438 (0.380, 0.496)
Renal hypodensities	122	0.613 (0.558, 0.669)	0.521 (0.467, 0.576)	0.603 (0.548, 0.658)	0.659 (0.605, 0.714)	0.545 (0.491, 0.600)	0.560 (0.505, 0.615)
Renal cyst	546	0.638 (0.610, 0.666)	0.559 (0.531, 0.588)	0.587 (0.559, 0.616)	0.646 (0.618, 0.674)	0.515 (0.487, 0.544)	0.642 (0.614, 0.670)
Hydronephrosis	112	0.605 (0.549, 0.662)	0.616 (0.559, 0.673)	0.482 (0.428, 0.537)	0.594 (0.537, 0.651)	0.570 (0.514, 0.627)	0.546 (0.490, 0.602)
Prostatomegaly	81	0.632 (0.565, 0.698)	0.632 (0.565, 0.699)	0.486 (0.421, 0.550)	0.551 (0.484, 0.617)	0.569 (0.502, 0.635)	0.510 (0.444, 0.575)
Metastatic disease	46	0.684 (0.589, 0.779)	0.793 (0.709, 0.876)	0.552 (0.453, 0.651)	0.603 (0.505, 0.702)	0.473 (0.376, 0.570)	0.543 (0.444, 0.642)
Lymphadenopathy	49	0.585 (0.493, 0.676)	0.655 (0.565, 0.745)	0.487 (0.397, 0.576)	0.600 (0.509, 0.692)	0.602 (0.510, 0.693)	0.490 (0.400, 0.580)
Anasarca	352	0.710 (0.678, 0.742)	0.874 (0.850, 0.898)	0.600 (0.567, 0.633)	0.303 (0.277, 0.330)	0.741 (0.710, 0.772)	0.492 (0.460, 0.525)

Table S5. Zero-shot finding prediction on RadChestCT. Class-wise AUROC values with 95% confidence intervals are reported for RadChestCT findings. **Bold** indicates the best result in each row.

Finding	N	TRACE	Merlin	CT-CLIP	MedSigLIP	BiomedCLIP	OpenAI CLIP
Cardiomegaly	394	0.700 (0.670, 0.730)	0.515 (0.485, 0.545)	0.614 (0.583, 0.645)	0.601 (0.570, 0.632)	0.631 (0.600, 0.662)	0.585 (0.554, 0.616)
Pericardial effusion	563	0.613 (0.586, 0.639)	0.496 (0.470, 0.521)	0.496 (0.470, 0.521)	0.523 (0.497, 0.550)	0.525 (0.499, 0.551)	0.537 (0.510, 0.563)
Arterial wall calcification	2635	0.711 (0.693, 0.728)	0.613 (0.594, 0.633)	0.531 (0.510, 0.552)	0.499 (0.478, 0.520)	0.595 (0.575, 0.615)	0.534 (0.513, 0.555)
Coronary artery calc.	1985	0.715 (0.698, 0.731)	0.612 (0.594, 0.630)	0.532 (0.513, 0.551)	0.529 (0.510, 0.547)	0.581 (0.563, 0.600)	0.524 (0.505, 0.542)
Pleural effusion	735	0.812 (0.793, 0.832)	0.672 (0.648, 0.695)	0.471 (0.448, 0.495)	0.478 (0.454, 0.501)	0.540 (0.517, 0.564)	0.608 (0.584, 0.631)
Peribronchial thick.	556	0.571 (0.544, 0.597)	0.558 (0.531, 0.584)	0.538 (0.512, 0.565)	0.520 (0.494, 0.547)	0.546 (0.520, 0.573)	0.546 (0.520, 0.573)
Bronchiectasis	597	0.652 (0.626, 0.677)	0.672 (0.647, 0.697)	0.503 (0.478, 0.529)	0.520 (0.495, 0.546)	0.548 (0.522, 0.574)	0.570 (0.544, 0.596)
Emphysema	992	0.641 (0.620, 0.662)	0.653 (0.632, 0.674)	0.545 (0.524, 0.566)	0.486 (0.465, 0.507)	0.486 (0.465, 0.507)	0.439 (0.418, 0.459)
Atelectasis	1335	0.574 (0.555, 0.594)	0.512 (0.493, 0.532)	0.501 (0.482, 0.521)	0.528 (0.509, 0.548)	0.432 (0.413, 0.451)	0.540 (0.520, 0.559)
Consolidation	505	0.735 (0.709, 0.761)	0.626 (0.598, 0.654)	0.478 (0.451, 0.505)	0.535 (0.507, 0.562)	0.476 (0.449, 0.503)	0.549 (0.521, 0.576)
Lung opacity	2847	0.664 (0.644, 0.684)	0.563 (0.541, 0.585)	0.520 (0.498, 0.543)	0.518 (0.496, 0.541)	0.562 (0.540, 0.584)	0.522 (0.499, 0.545)
Lung nodule	2937	0.610 (0.588, 0.632)	0.602 (0.580, 0.625)	0.531 (0.508, 0.555)	0.492 (0.468, 0.516)	0.436 (0.412, 0.460)	0.490 (0.466, 0.514)
Pulm. fibrotic sequela	1391	0.605 (0.586, 0.624)	0.618 (0.599, 0.637)	0.502 (0.482, 0.521)	0.490 (0.471, 0.510)	0.540 (0.521, 0.559)	0.524 (0.505, 0.543)
Mosaic attenuation	118	0.603 (0.549, 0.658)	0.530 (0.476, 0.584)	0.525 (0.471, 0.578)	0.532 (0.478, 0.586)	0.387 (0.340, 0.435)	0.388 (0.340, 0.435)
Interlobular septal thick.	252	0.730 (0.694, 0.767)	0.617 (0.579, 0.655)	0.541 (0.503, 0.579)	0.552 (0.514, 0.590)	0.697 (0.660, 0.734)	0.540 (0.502, 0.577)
Medical material	2217	0.558 (0.539, 0.577)	0.550 (0.531, 0.569)	0.529 (0.510, 0.548)	0.525 (0.506, 0.545)	0.551 (0.532, 0.570)	0.521 (0.502, 0.540)
Lymphadenopathy	598	0.677 (0.651, 0.702)	0.520 (0.494, 0.545)	0.544 (0.518, 0.570)	0.498 (0.472, 0.523)	0.502 (0.477, 0.527)	0.573 (0.547, 0.599)
Hiatal hernia	425	0.587 (0.557, 0.617)	0.562 (0.532, 0.591)	0.528 (0.498, 0.557)	0.524 (0.495, 0.554)	0.547 (0.517, 0.577)	0.583 (0.553, 0.613)

Table S6. Linear-probe phenotype prediction on INSPECT, part I. Class-wise AUROC values with 95% confidence intervals are reported for the first 49 INSPECT phenotypes after supervised linear probing. **Bold** indicates the best result in each row.

Finding	N	TRACE	Merlin	CT-CLIP	MedSigLIP	BiomedCLIP	OpenAI CLIP
Heart valve dis.	95	0.737 (0.679, 0.795)	0.487 (0.428, 0.545)	0.559 (0.498, 0.619)	0.533 (0.473, 0.593)	0.587 (0.526, 0.648)	0.559 (0.498, 0.619)
Hypotension	229	0.752 (0.714, 0.789)	0.512 (0.473, 0.551)	0.526 (0.487, 0.566)	0.586 (0.546, 0.626)	0.603 (0.563, 0.643)	0.526 (0.487, 0.566)
Shock	86	0.799 (0.742, 0.856)	0.536 (0.473, 0.599)	0.539 (0.476, 0.602)	0.543 (0.480, 0.606)	0.576 (0.512, 0.640)	0.539 (0.476, 0.602)
Hypertension	847	0.714 (0.692, 0.735)	0.573 (0.550, 0.596)	0.515 (0.492, 0.538)	0.521 (0.498, 0.544)	0.522 (0.499, 0.545)	0.515 (0.492, 0.538)
Cardiomegaly	228	0.687 (0.647, 0.726)	0.476 (0.437, 0.514)	0.515 (0.476, 0.554)	0.571 (0.531, 0.611)	0.574 (0.534, 0.614)	0.515 (0.476, 0.554)
Cardiac conduction dis.	981	0.653 (0.632, 0.675)	0.569 (0.547, 0.591)	0.519 (0.497, 0.541)	0.552 (0.530, 0.574)	0.569 (0.547, 0.591)	0.519 (0.497, 0.541)
Rheumatic valve dis.	62	0.632 (0.557, 0.707)	0.407 (0.341, 0.474)	0.470 (0.399, 0.541)	0.419 (0.351, 0.486)	0.480 (0.408, 0.551)	0.470 (0.399, 0.541)
Pulmonary heart dis.	459	0.601 (0.572, 0.631)	0.513 (0.484, 0.542)	0.508 (0.480, 0.537)	0.551 (0.522, 0.581)	0.558 (0.529, 0.587)	0.508 (0.480, 0.537)
Cardiac/circulatory anomalies	56	0.621 (0.542, 0.700)	0.473 (0.398, 0.547)	0.457 (0.383, 0.531)	0.595 (0.516, 0.674)	0.552 (0.473, 0.630)	0.457 (0.383, 0.531)
Other pulm. inflam./edema	88	0.822 (0.768, 0.876)	0.475 (0.415, 0.535)	0.497 (0.436, 0.558)	0.551 (0.488, 0.614)	0.603 (0.540, 0.666)	0.497 (0.436, 0.558)
Pneumonia	342	0.818 (0.790, 0.846)	0.469 (0.437, 0.500)	0.499 (0.467, 0.532)	0.541 (0.508, 0.574)	0.584 (0.550, 0.617)	0.499 (0.467, 0.532)
Empyema/pneumothorax	92	0.824 (0.771, 0.876)	0.549 (0.488, 0.611)	0.502 (0.442, 0.562)	0.589 (0.527, 0.651)	0.616 (0.554, 0.678)	0.502 (0.442, 0.562)
Aspiration pneumonitis	123	0.781 (0.732, 0.830)	0.454 (0.404, 0.505)	0.486 (0.434, 0.537)	0.555 (0.502, 0.608)	0.583 (0.529, 0.637)	0.486 (0.434, 0.537)
Pulm. congestion/hypostasis	207	0.812 (0.776, 0.848)	0.490 (0.450, 0.531)	0.557 (0.515, 0.598)	0.597 (0.555, 0.639)	0.638 (0.597, 0.680)	0.557 (0.515, 0.598)
Other respiratory dis.	79	0.742 (0.679, 0.806)	0.498 (0.433, 0.562)	0.456 (0.393, 0.518)	0.549 (0.483, 0.615)	0.551 (0.485, 0.618)	0.456 (0.393, 0.518)
Asphyxia/hypoxemia	357	0.745 (0.714, 0.775)	0.468 (0.437, 0.499)	0.497 (0.465, 0.529)	0.530 (0.498, 0.563)	0.559 (0.526, 0.591)	0.497 (0.465, 0.529)
Pleurisy/pleural effusion	576	0.841 (0.820, 0.862)	0.561 (0.535, 0.588)	0.504 (0.478, 0.530)	0.563 (0.537, 0.590)	0.609 (0.582, 0.635)	0.504 (0.478, 0.530)
Chronic airway obstruction	261	0.754 (0.719, 0.789)	0.506 (0.469, 0.543)	0.484 (0.448, 0.520)	0.543 (0.505, 0.580)	0.537 (0.500, 0.575)	0.484 (0.448, 0.520)
Abnormal lung findings	957	0.686 (0.665, 0.708)	0.569 (0.547, 0.591)	0.499 (0.477, 0.521)	0.541 (0.519, 0.563)	0.557 (0.535, 0.579)	0.499 (0.477, 0.521)
Pulm. collapse/emphysema	342	0.757 (0.726, 0.788)	0.541 (0.508, 0.574)	0.522 (0.489, 0.554)	0.559 (0.526, 0.592)	0.594 (0.561, 0.628)	0.522 (0.489, 0.554)
Other lung dis.	144	0.649 (0.600, 0.699)	0.569 (0.520, 0.619)	0.485 (0.437, 0.532)	0.562 (0.513, 0.612)	0.543 (0.494, 0.592)	0.485 (0.437, 0.532)
Abnormal sputum	58	0.708 (0.633, 0.784)	0.528 (0.452, 0.604)	0.519 (0.443, 0.595)	0.608 (0.530, 0.686)	0.597 (0.519, 0.675)	0.519 (0.443, 0.595)
Respiratory abnormalities	93	0.670 (0.609, 0.731)	0.476 (0.418, 0.535)	0.545 (0.484, 0.606)	0.531 (0.470, 0.591)	0.563 (0.502, 0.624)	0.545 (0.484, 0.606)
Other upper resp. dis.	50	0.616 (0.533, 0.700)	0.538 (0.456, 0.620)	0.462 (0.384, 0.540)	0.539 (0.457, 0.622)	0.519 (0.437, 0.601)	0.462 (0.384, 0.540)
Asthma	189	0.588 (0.544, 0.632)	0.496 (0.454, 0.538)	0.506 (0.463, 0.548)	0.522 (0.479, 0.565)	0.516 (0.473, 0.559)	0.506 (0.463, 0.548)
Venous embolism/thrombosis	235	0.657 (0.618, 0.696)	0.529 (0.490, 0.568)	0.521 (0.482, 0.559)	0.543 (0.504, 0.582)	0.541 (0.501, 0.580)	0.521 (0.482, 0.559)
Cerebrovascular dis.	108	0.676 (0.619, 0.733)	0.473 (0.419, 0.528)	0.498 (0.443, 0.553)	0.545 (0.488, 0.601)	0.532 (0.476, 0.589)	0.498 (0.443, 0.553)
Other aneurysm	55	0.671 (0.592, 0.751)	0.530 (0.452, 0.608)	0.445 (0.371, 0.518)	0.555 (0.476, 0.634)	0.546 (0.467, 0.624)	0.445 (0.371, 0.518)
Prostate hyperplasia	85	0.735 (0.673, 0.796)	0.501 (0.439, 0.563)	0.535 (0.471, 0.598)	0.507 (0.445, 0.570)	0.543 (0.479, 0.607)	0.535 (0.471, 0.598)
Other urinary disorders	156	0.708 (0.662, 0.755)	0.468 (0.423, 0.514)	0.504 (0.457, 0.550)	0.545 (0.497, 0.592)	0.546 (0.499, 0.594)	0.504 (0.457, 0.550)
Kidney/ureter disorders	69	0.692 (0.622, 0.762)	0.517 (0.447, 0.586)	0.476 (0.408, 0.543)	0.534 (0.464, 0.604)	0.528 (0.458, 0.598)	0.476 (0.408, 0.543)
Protein-calorie malnutrition	271	0.772 (0.739, 0.806)	0.544 (0.508, 0.581)	0.487 (0.451, 0.522)	0.587 (0.550, 0.624)	0.589 (0.552, 0.626)	0.487 (0.451, 0.522)
Diabetes mellitus	434	0.715 (0.686, 0.743)	0.484 (0.455, 0.513)	0.535 (0.505, 0.564)	0.547 (0.518, 0.577)	0.551 (0.521, 0.581)	0.535 (0.505, 0.564)
Mineral metabolism dis.	139	0.697 (0.647, 0.747)	0.458 (0.411, 0.506)	0.474 (0.425, 0.522)	0.473 (0.425, 0.521)	0.521 (0.472, 0.571)	0.474 (0.425, 0.522)
Obesity/overnutrition	224	0.806 (0.771, 0.841)	0.469 (0.431, 0.508)	0.570 (0.530, 0.610)	0.570 (0.530, 0.611)	0.528 (0.488, 0.568)	0.570 (0.530, 0.610)
Ascites	122	0.821 (0.775, 0.867)	0.564 (0.510, 0.618)	0.541 (0.488, 0.594)	0.562 (0.508, 0.615)	0.565 (0.511, 0.619)	0.541 (0.488, 0.594)
Other peritoneal dis.	59	0.798 (0.729, 0.866)	0.543 (0.466, 0.619)	0.473 (0.400, 0.546)	0.574 (0.497, 0.650)	0.589 (0.512, 0.666)	0.473 (0.400, 0.546)
Dysphagia	138	0.733 (0.685, 0.782)	0.457 (0.410, 0.505)	0.469 (0.421, 0.517)	0.522 (0.472, 0.572)	0.568 (0.517, 0.619)	0.469 (0.421, 0.517)
Digestive symptoms	193	0.724 (0.683, 0.766)	0.479 (0.438, 0.521)	0.491 (0.449, 0.533)	0.518 (0.476, 0.561)	0.558 (0.515, 0.601)	0.491 (0.449, 0.533)
Pancreatic diseases	58	0.668 (0.591, 0.745)	0.525 (0.449, 0.601)	0.541 (0.464, 0.618)	0.534 (0.458, 0.611)	0.550 (0.473, 0.627)	0.541 (0.464, 0.618)
Other intestinal dis.	111	0.784 (0.733, 0.835)	0.513 (0.457, 0.568)	0.513 (0.458, 0.568)	0.609 (0.552, 0.665)	0.604 (0.548, 0.661)	0.513 (0.458, 0.568)
Abdominal hernia	88	0.701 (0.639, 0.763)	0.508 (0.446, 0.569)	0.564 (0.501, 0.627)	0.512 (0.450, 0.573)	0.585 (0.522, 0.649)	0.564 (0.501, 0.627)
Constipation	201	0.723 (0.682, 0.764)	0.545 (0.503, 0.588)	0.522 (0.481, 0.564)	0.582 (0.539, 0.624)	0.597 (0.554, 0.639)	0.522 (0.481, 0.564)
Esophageal diseases	342	0.634 (0.601, 0.667)	0.491 (0.459, 0.523)	0.507 (0.474, 0.539)	0.518 (0.485, 0.551)	0.515 (0.482, 0.547)	0.507 (0.474, 0.539)
Functional digestive dis.	81	0.657 (0.591, 0.723)	0.514 (0.449, 0.578)	0.521 (0.456, 0.585)	0.500 (0.436, 0.564)	0.551 (0.486, 0.616)	0.521 (0.456, 0.585)
Other liver dis.	212	0.677 (0.636, 0.717)	0.527 (0.486, 0.568)	0.522 (0.481, 0.563)	0.546 (0.505, 0.587)	0.568 (0.526, 0.609)	0.522 (0.481, 0.563)
Nausea/vomiting	188	0.614 (0.570, 0.658)	0.514 (0.471, 0.557)	0.480 (0.438, 0.522)	0.531 (0.488, 0.574)	0.536 (0.493, 0.580)	0.480 (0.438, 0.522)
Abdominal pain	274	0.637 (0.601, 0.674)	0.491 (0.456, 0.527)	0.499 (0.464, 0.535)	0.545 (0.508, 0.581)	0.523 (0.487, 0.559)	0.499 (0.464, 0.535)
White blood cell dis.	194	0.718 (0.676, 0.759)	0.451 (0.411, 0.492)	0.526 (0.484, 0.569)	0.523 (0.480, 0.565)	0.566 (0.523, 0.609)	0.526 (0.484, 0.569)

Table S7. Linear-probe phenotype prediction on INSPECT, part II. Class-wise AUROC values with 95% confidence intervals are reported for the remaining 48 INSPECT phenotypes after supervised linear probing. **Bold** indicates the best result in each row.

Finding	N	TRACE	Merlin	CT-CLIP	MedSigLIP	BiomedCLIP	OpenAI CLIP
Other anemias	473	0.731 (0.704, 0.759)	0.550 (0.521, 0.578)	0.488 (0.460, 0.516)	0.545 (0.516, 0.574)	0.537 (0.509, 0.566)	0.488 (0.460, 0.516)
Purpura/hemorrhagic dis.	134	0.680 (0.629, 0.731)	0.509 (0.458, 0.559)	0.495 (0.445, 0.545)	0.494 (0.445, 0.544)	0.509 (0.459, 0.559)	0.495 (0.445, 0.545)
Aplastic anemia	78	0.701 (0.635, 0.766)	0.516 (0.450, 0.581)	0.517 (0.452, 0.583)	0.510 (0.445, 0.576)	0.575 (0.508, 0.642)	0.517 (0.452, 0.583)
Other hematologic dis.	171	0.597 (0.551, 0.643)	0.555 (0.509, 0.600)	0.461 (0.418, 0.504)	0.528 (0.483, 0.573)	0.509 (0.464, 0.554)	0.461 (0.418, 0.504)
Nonspecific blood findings	186	0.575 (0.531, 0.619)	0.443 (0.402, 0.484)	0.487 (0.444, 0.529)	0.480 (0.438, 0.522)	0.471 (0.429, 0.513)	0.487 (0.444, 0.529)
Cellulitis/abscess	55	0.618 (0.538, 0.698)	0.526 (0.448, 0.605)	0.518 (0.440, 0.596)	0.573 (0.493, 0.652)	0.576 (0.496, 0.656)	0.518 (0.440, 0.596)
Urinary tract infection	90	0.686 (0.624, 0.748)	0.471 (0.411, 0.530)	0.508 (0.447, 0.569)	0.536 (0.474, 0.598)	0.520 (0.459, 0.581)	0.508 (0.447, 0.569)
Respiratory cancer	145	0.838 (0.798, 0.879)	0.567 (0.518, 0.617)	0.451 (0.405, 0.497)	0.546 (0.497, 0.595)	0.547 (0.498, 0.597)	0.451 (0.405, 0.497)
Suspected/other cancer	179	0.750 (0.708, 0.792)	0.554 (0.509, 0.598)	0.497 (0.453, 0.540)	0.570 (0.525, 0.615)	0.578 (0.533, 0.623)	0.497 (0.453, 0.540)
Leukemia	62	0.531 (0.457, 0.604)	0.518 (0.444, 0.591)	0.555 (0.480, 0.630)	0.530 (0.457, 0.604)	0.574 (0.499, 0.649)	0.555 (0.480, 0.630)
Secondary malignancy	372	0.755 (0.726, 0.785)	0.564 (0.532, 0.596)	0.479 (0.448, 0.510)	0.542 (0.510, 0.574)	0.564 (0.532, 0.596)	0.479 (0.448, 0.510)
Breast cancer	90	0.583 (0.520, 0.645)	0.551 (0.489, 0.613)	0.462 (0.403, 0.520)	0.458 (0.400, 0.517)	0.482 (0.422, 0.542)	0.462 (0.403, 0.520)
Neurological dis.	200	0.730 (0.690, 0.771)	0.483 (0.442, 0.524)	0.508 (0.467, 0.550)	0.564 (0.521, 0.606)	0.575 (0.533, 0.618)	0.508 (0.467, 0.550)
Dementia/cognitive dis.	158	0.769 (0.725, 0.812)	0.478 (0.432, 0.523)	0.518 (0.471, 0.564)	0.567 (0.520, 0.615)	0.594 (0.546, 0.641)	0.518 (0.471, 0.564)
Inflammatory/toxic neuropathy	51	0.614 (0.531, 0.697)	0.378 (0.307, 0.448)	0.525 (0.444, 0.606)	0.487 (0.408, 0.566)	0.512 (0.431, 0.592)	0.525 (0.444, 0.606)
Other brain conditions	140	0.726 (0.678, 0.775)	0.497 (0.448, 0.546)	0.553 (0.503, 0.603)	0.555 (0.505, 0.605)	0.603 (0.553, 0.654)	0.553 (0.503, 0.603)
Sleep dis.	295	0.695 (0.661, 0.730)	0.452 (0.419, 0.486)	0.501 (0.466, 0.535)	0.528 (0.493, 0.563)	0.525 (0.490, 0.560)	0.501 (0.466, 0.535)
Anxiety dis.	270	0.658 (0.622, 0.695)	0.466 (0.431, 0.501)	0.500 (0.464, 0.536)	0.505 (0.469, 0.542)	0.539 (0.502, 0.575)	0.500 (0.464, 0.536)
Adjustment reaction	64	0.742 (0.672, 0.813)	0.487 (0.416, 0.558)	0.510 (0.438, 0.582)	0.572 (0.499, 0.646)	0.577 (0.503, 0.651)	0.510 (0.438, 0.582)
Epilepsy/seizures	74	0.661 (0.592, 0.729)	0.473 (0.408, 0.539)	0.519 (0.451, 0.586)	0.535 (0.467, 0.603)	0.562 (0.494, 0.631)	0.519 (0.451, 0.586)
Mood dis.	242	0.612 (0.573, 0.651)	0.484 (0.447, 0.522)	0.495 (0.457, 0.533)	0.531 (0.493, 0.570)	0.526 (0.488, 0.564)	0.495 (0.457, 0.533)
Substance addiction/dis.	81	0.632 (0.566, 0.698)	0.465 (0.402, 0.527)	0.533 (0.468, 0.598)	0.556 (0.491, 0.622)	0.581 (0.515, 0.647)	0.533 (0.468, 0.598)
Tobacco use dis.	125	0.570 (0.517, 0.623)	0.469 (0.419, 0.520)	0.471 (0.421, 0.522)	0.489 (0.438, 0.541)	0.506 (0.454, 0.557)	0.471 (0.421, 0.522)
Alcohol-related dis.	86	0.666 (0.602, 0.729)	0.527 (0.464, 0.590)	0.487 (0.426, 0.548)	0.470 (0.410, 0.531)	0.534 (0.471, 0.597)	0.487 (0.426, 0.548)
Other headache syndromes	50	0.569 (0.485, 0.652)	0.531 (0.448, 0.613)	0.532 (0.450, 0.614)	0.613 (0.529, 0.697)	0.592 (0.509, 0.676)	0.532 (0.450, 0.614)
Vestibular dis.	94	0.542 (0.481, 0.602)	0.488 (0.429, 0.547)	0.466 (0.409, 0.524)	0.488 (0.429, 0.547)	0.442 (0.385, 0.499)	0.466 (0.409, 0.524)
Transplant complications	126	0.727 (0.676, 0.778)	0.535 (0.483, 0.587)	0.532 (0.479, 0.584)	0.610 (0.557, 0.664)	0.599 (0.545, 0.652)	0.532 (0.479, 0.584)
Muscle symptoms	74	0.669 (0.600, 0.737)	0.503 (0.436, 0.569)	0.500 (0.433, 0.566)	0.567 (0.499, 0.636)	0.612 (0.543, 0.681)	0.500 (0.433, 0.566)
MSK nonspecific findings	53	0.643 (0.562, 0.724)	0.449 (0.374, 0.524)	0.509 (0.430, 0.588)	0.526 (0.446, 0.605)	0.502 (0.423, 0.580)	0.509 (0.430, 0.588)
Osteoarthritis	85	0.686 (0.623, 0.750)	0.472 (0.411, 0.533)	0.497 (0.435, 0.559)	0.519 (0.456, 0.582)	0.501 (0.439, 0.563)	0.497 (0.435, 0.559)
Back pain	124	0.578 (0.525, 0.632)	0.465 (0.414, 0.515)	0.492 (0.440, 0.543)	0.489 (0.437, 0.540)	0.484 (0.432, 0.535)	0.492 (0.440, 0.543)
Joint pain	108	0.592 (0.535, 0.649)	0.497 (0.442, 0.553)	0.543 (0.487, 0.600)	0.520 (0.464, 0.576)	0.507 (0.451, 0.562)	0.543 (0.487, 0.600)
Injury	58	0.615 (0.537, 0.693)	0.489 (0.415, 0.564)	0.523 (0.447, 0.599)	0.473 (0.399, 0.546)	0.544 (0.468, 0.621)	0.523 (0.447, 0.599)
Limb pain	99	0.553 (0.493, 0.612)	0.540 (0.481, 0.599)	0.500 (0.443, 0.558)	0.513 (0.454, 0.571)	0.496 (0.438, 0.554)	0.500 (0.443, 0.558)
Limb MSK symptoms	121	0.586 (0.531, 0.640)	0.535 (0.482, 0.589)	0.499 (0.446, 0.551)	0.538 (0.485, 0.592)	0.528 (0.475, 0.581)	0.499 (0.446, 0.551)
Other soft tissue dis.	108	0.598 (0.540, 0.655)	0.527 (0.471, 0.584)	0.531 (0.475, 0.588)	0.534 (0.478, 0.590)	0.536 (0.480, 0.593)	0.531 (0.475, 0.588)
Immune disorders	68	0.745 (0.677, 0.813)	0.519 (0.449, 0.589)	0.510 (0.440, 0.580)	0.568 (0.496, 0.640)	0.615 (0.543, 0.687)	0.510 (0.440, 0.580)
Procedure complications	58	0.760 (0.688, 0.833)	0.564 (0.487, 0.642)	0.558 (0.481, 0.635)	0.593 (0.515, 0.670)	0.636 (0.558, 0.713)	0.558 (0.481, 0.635)
Ill-defined morbidity	60	0.746 (0.673, 0.818)	0.511 (0.437, 0.586)	0.442 (0.371, 0.512)	0.509 (0.435, 0.583)	0.470 (0.398, 0.542)	0.442 (0.371, 0.512)
Malaise/fatigue	307	0.699 (0.665, 0.732)	0.510 (0.476, 0.544)	0.496 (0.462, 0.530)	0.546 (0.512, 0.581)	0.560 (0.525, 0.595)	0.496 (0.462, 0.530)
Fever unknown origin	209	0.667 (0.625, 0.708)	0.497 (0.456, 0.537)	0.498 (0.457, 0.538)	0.517 (0.476, 0.558)	0.540 (0.499, 0.582)	0.498 (0.457, 0.538)
Acquired organ absence	163	0.660 (0.613, 0.707)	0.472 (0.427, 0.517)	0.517 (0.471, 0.563)	0.522 (0.476, 0.568)	0.550 (0.503, 0.597)	0.517 (0.471, 0.563)
Toxic solvent/alcohol effect	135	0.706 (0.655, 0.756)	0.502 (0.452, 0.552)	0.524 (0.474, 0.575)	0.544 (0.494, 0.595)	0.599 (0.547, 0.650)	0.524 (0.474, 0.575)
Skin symptoms	90	0.586 (0.523, 0.648)	0.459 (0.401, 0.518)	0.471 (0.412, 0.530)	0.507 (0.447, 0.568)	0.492 (0.431, 0.552)	0.471 (0.412, 0.530)
Abdomen/pelvis symptoms	65	0.629 (0.555, 0.702)	0.510 (0.438, 0.581)	0.504 (0.433, 0.575)	0.509 (0.438, 0.581)	0.526 (0.454, 0.598)	0.504 (0.433, 0.575)
Other tests	97	0.688 (0.628, 0.747)	0.499 (0.441, 0.558)	0.542 (0.482, 0.601)	0.538 (0.479, 0.598)	0.558 (0.498, 0.618)	0.542 (0.482, 0.601)
Syncope/collapse	98	0.574 (0.514, 0.634)	0.497 (0.439, 0.555)	0.519 (0.460, 0.578)	0.450 (0.394, 0.506)	0.506 (0.448, 0.564)	0.519 (0.460, 0.578)
Nonspecific chest pain	701	0.636 (0.612, 0.661)	0.476 (0.452, 0.500)	0.498 (0.473, 0.522)	0.468 (0.444, 0.492)	0.465 (0.441, 0.489)	0.498 (0.473, 0.522)

Table S8. Linear-probe finding prediction on CT-RATE. Class-wise AUROC values with 95% confidence intervals are reported for the 18 CT-RATE findings after supervised linear probing. **Bold** indicates the best result in each row.

Finding	N	TRACE	Merlin	CT-CLIP	MedSigLIP	BiomedCLIP	OpenAI CLIP
Cardiomegaly	273	0.898 (0.873, 0.923)	0.674 (0.637, 0.710)	0.876 (0.849, 0.903)	0.728 (0.692, 0.763)	0.640 (0.603, 0.677)	0.645 (0.608, 0.682)
Pericardial effusion	205	0.849 (0.815, 0.882)	0.652 (0.610, 0.695)	0.773 (0.734, 0.812)	0.617 (0.575, 0.660)	0.527 (0.485, 0.568)	0.614 (0.572, 0.657)
Arterial wall calcification	733	0.883 (0.866, 0.899)	0.666 (0.642, 0.690)	0.853 (0.835, 0.872)	0.795 (0.774, 0.816)	0.662 (0.638, 0.686)	0.611 (0.586, 0.636)
Coronary artery calc.	643	0.875 (0.856, 0.893)	0.637 (0.611, 0.662)	0.844 (0.824, 0.864)	0.762 (0.739, 0.785)	0.659 (0.633, 0.684)	0.586 (0.560, 0.612)
Pleural effusion	336	0.949 (0.932, 0.966)	0.690 (0.657, 0.723)	0.905 (0.883, 0.927)	0.712 (0.679, 0.744)	0.551 (0.517, 0.584)	0.674 (0.641, 0.708)
Peribronchial thick.	312	0.780 (0.749, 0.812)	0.568 (0.533, 0.603)	0.715 (0.681, 0.748)	0.519 (0.485, 0.554)	0.520 (0.486, 0.555)	0.503 (0.468, 0.537)
Bronchiectasis	267	0.779 (0.746, 0.813)	0.548 (0.511, 0.585)	0.640 (0.603, 0.678)	0.547 (0.510, 0.585)	0.530 (0.493, 0.567)	0.431 (0.396, 0.466)
Emphysema	496	0.786 (0.761, 0.811)	0.541 (0.513, 0.570)	0.727 (0.700, 0.754)	0.553 (0.525, 0.582)	0.544 (0.515, 0.572)	0.511 (0.482, 0.539)
Atelectasis	600	0.741 (0.716, 0.765)	0.584 (0.558, 0.611)	0.679 (0.653, 0.705)	0.640 (0.613, 0.666)	0.579 (0.552, 0.606)	0.563 (0.537, 0.590)
Consolidation	490	0.878 (0.857, 0.898)	0.570 (0.541, 0.599)	0.729 (0.701, 0.756)	0.589 (0.560, 0.618)	0.486 (0.458, 0.515)	0.578 (0.549, 0.607)
Lung opacity	971	0.839 (0.822, 0.856)	0.567 (0.544, 0.590)	0.653 (0.631, 0.676)	0.573 (0.550, 0.596)	0.532 (0.508, 0.555)	0.549 (0.526, 0.573)
Lung nodule	1123	0.689 (0.668, 0.710)	0.511 (0.489, 0.534)	0.645 (0.624, 0.667)	0.513 (0.490, 0.535)	0.528 (0.506, 0.551)	0.553 (0.530, 0.576)
Pulm. fibrotic sequela	673	0.699 (0.674, 0.723)	0.519 (0.493, 0.544)	0.638 (0.613, 0.664)	0.529 (0.503, 0.554)	0.536 (0.510, 0.562)	0.468 (0.443, 0.493)
Mosaic attenuation	213	0.817 (0.782, 0.853)	0.564 (0.523, 0.606)	0.729 (0.689, 0.768)	0.564 (0.523, 0.606)	0.620 (0.578, 0.661)	0.588 (0.547, 0.630)
Interlobular septal thick.	206	0.857 (0.824, 0.889)	0.609 (0.567, 0.652)	0.800 (0.763, 0.837)	0.616 (0.574, 0.658)	0.560 (0.518, 0.602)	0.594 (0.551, 0.636)
Medical material	277	0.777 (0.744, 0.811)	0.629 (0.592, 0.666)	0.706 (0.670, 0.741)	0.658 (0.622, 0.695)	0.539 (0.502, 0.575)	0.597 (0.560, 0.634)
Lymphadenopathy	693	0.723 (0.699, 0.746)	0.590 (0.564, 0.615)	0.714 (0.690, 0.737)	0.584 (0.558, 0.609)	0.549 (0.524, 0.575)	0.567 (0.541, 0.592)
Hiatal hernia	341	0.736 (0.704, 0.767)	0.524 (0.490, 0.557)	0.706 (0.674, 0.738)	0.531 (0.498, 0.565)	0.591 (0.557, 0.624)	0.517 (0.484, 0.550)

Table S9. Concept-model comparison on INSPECT, part I. AUROC values and LLM interpretability scores compare TRACE and CBM variants using refined or full concept banks. **Bold** indicates the best result in each row.

Finding	N	TRACE Refined set		TRACE Full set		CBM Refined set		CBM Full set	
		AUROC	LLM	AUROC	LLM	AUROC	LLM	AUROC	LLM
Pleurisy/pleural effusion	576	0.841 $\pm$ 0.001	3.50	0.842 $\pm$ 0.000	4.55	0.844 $\pm$ 0.001	3.15	0.799 $\pm$ 0.088	3.15
Respiratory cancer	145	0.838 $\pm$ 0.005	2.45	0.834 $\pm$ 0.006	1.70	0.844 $\pm$ 0.009	2.80	0.753 $\pm$ 0.039	2.80
Empyema/pneumothorax	92	0.824 $\pm$ 0.003	2.85	0.826 $\pm$ 0.003	1.60	0.826 $\pm$ 0.017	2.50	0.690 $\pm$ 0.177	2.50
Other pulm. inflam./edema	88	0.822 $\pm$ 0.010	2.55	0.815 $\pm$ 0.006	1.75	0.820 $\pm$ 0.011	2.25	0.730 $\pm$ 0.132	2.25
Ascites	122	0.821 $\pm$ 0.006	2.00	0.823 $\pm$ 0.001	1.00	0.824 $\pm$ 0.010	2.70	0.751 $\pm$ 0.111	2.70
Pneumonia	342	0.818 $\pm$ 0.002	3.60	0.816 $\pm$ 0.002	3.90	0.819 $\pm$ 0.002	1.15	0.722 $\pm$ 0.084	1.15
Pulm. congestion/hypostasis	207	0.812 $\pm$ 0.003	2.70	0.812 $\pm$ 0.002	3.20	0.812 $\pm$ 0.005	2.00	0.767 $\pm$ 0.062	2.00
Obesity/overnutrition	224	0.806 $\pm$ 0.005	1.85	0.803 $\pm$ 0.009	1.15	0.815 $\pm$ 0.005	1.30	0.649 $\pm$ 0.067	1.30
Shock	86	0.799 $\pm$ 0.007	2.50	0.800 $\pm$ 0.003	1.05	0.802 $\pm$ 0.006	2.15	0.660 $\pm$ 0.156	2.15
Other peritoneal dis.	59	0.798 $\pm$ 0.010	1.60	0.812 $\pm$ 0.004	1.00	0.816 $\pm$ 0.019	1.65	0.704 $\pm$ 0.130	1.65
Other intestinal dis.	111	0.784 $\pm$ 0.004	1.00	0.795 $\pm$ 0.005	2.00	0.796 $\pm$ 0.009	1.90	0.732 $\pm$ 0.123	1.90
Aspiration pneumonitis	123	0.781 $\pm$ 0.005	2.00	0.779 $\pm$ 0.005	1.60	0.778 $\pm$ 0.012	1.55	0.627 $\pm$ 0.140	1.55
Protein-calorie malnutrition	271	0.772 $\pm$ 0.003	3.55	0.778 $\pm$ 0.002	2.20	0.778 $\pm$ 0.003	1.25	0.705 $\pm$ 0.095	1.25
Dementia/cognitive dis.	158	0.769 $\pm$ 0.006	1.65	0.775 $\pm$ 0.003	1.05	0.774 $\pm$ 0.007	1.45	0.672 $\pm$ 0.123	1.45
Procedure complications	58	0.760 $\pm$ 0.015	2.20	0.766 $\pm$ 0.005	2.10	0.764 $\pm$ 0.028	2.65	0.670 $\pm$ 0.128	2.65
Pulm. collapse/emphysema	342	0.757 $\pm$ 0.002	3.15	0.759 $\pm$ 0.001	2.60	0.760 $\pm$ 0.004	1.65	0.706 $\pm$ 0.097	1.65
Secondary malignancy	372	0.755 $\pm$ 0.002	2.00	0.750 $\pm$ 0.002	2.05	0.758 $\pm$ 0.004	3.30	0.717 $\pm$ 0.022	3.30
Chronic airway obstruction	261	0.754 $\pm$ 0.003	3.70	0.765 $\pm$ 0.004	3.50	0.761 $\pm$ 0.004	1.60	0.665 $\pm$ 0.042	1.60
Hypotension	229	0.752 $\pm$ 0.002	2.00	0.748 $\pm$ 0.001	1.00	0.747 $\pm$ 0.008	1.40	0.662 $\pm$ 0.139	1.40
Suspected/other cancer	179	0.750 $\pm$ 0.001	3.10	0.748 $\pm$ 0.001	2.70	0.748 $\pm$ 0.005	3.20	0.702 $\pm$ 0.051	3.20
Ill-defined morbidity	60	0.746 $\pm$ 0.007	1.95	0.746 $\pm$ 0.003	3.65	0.737 $\pm$ 0.027	1.45	0.666 $\pm$ 0.096	1.45
Immune disorders	68	0.745 $\pm$ 0.012	2.30	0.753 $\pm$ 0.005	2.10	0.745 $\pm$ 0.017	1.45	0.686 $\pm$ 0.085	1.45
Asphyxia/hypoxemia	357	0.745 $\pm$ 0.003	2.30	0.746 $\pm$ 0.002	1.70	0.749 $\pm$ 0.003	1.45	0.661 $\pm$ 0.101	1.45
Adjustment reaction	64	0.742 $\pm$ 0.009	1.95	0.748 $\pm$ 0.004	1.55	0.740 $\pm$ 0.024	1.00	0.660 $\pm$ 0.122	1.00
Other respiratory dis.	79	0.742 $\pm$ 0.005	2.50	0.753 $\pm$ 0.002	2.35	0.748 $\pm$ 0.014	1.00	0.711 $\pm$ 0.052	1.00
Heart valve dis.	95	0.737 $\pm$ 0.010	2.95	0.737 $\pm$ 0.004	2.35	0.735 $\pm$ 0.009	1.15	0.634 $\pm$ 0.105	1.15
Prostate hyperplasia	85	0.735 $\pm$ 0.012	1.60	0.722 $\pm$ 0.007	1.10	0.720 $\pm$ 0.013	1.05	0.668 $\pm$ 0.049	1.05
Dysphagia	138	0.733 $\pm$ 0.011	2.00	0.739 $\pm$ 0.008	1.80	0.741 $\pm$ 0.014	1.30	0.641 $\pm$ 0.118	1.30
Other anemias	473	0.731 $\pm$ 0.002	3.35	0.738 $\pm$ 0.001	2.00	0.739 $\pm$ 0.002	1.65	0.640 $\pm$ 0.141	1.65
Neurological dis.	200	0.730 $\pm$ 0.005	1.90	0.741 $\pm$ 0.004	1.10	0.747 $\pm$ 0.005	1.20	0.650 $\pm$ 0.111	1.20
Transplant complications	126	0.727 $\pm$ 0.009	2.15	0.725 $\pm$ 0.005	1.85	0.718 $\pm$ 0.006	1.55	0.633 $\pm$ 0.135	1.55
Other brain conditions	140	0.726 $\pm$ 0.009	1.95	0.732 $\pm$ 0.004	2.70	0.738 $\pm$ 0.009	1.55	0.655 $\pm$ 0.115	1.55
Digestive symptoms	193	0.724 $\pm$ 0.003	1.80	0.728 $\pm$ 0.002	1.95	0.737 $\pm$ 0.008	2.00	0.680 $\pm$ 0.063	2.00
Constipation	201	0.723 $\pm$ 0.002	1.95	0.726 $\pm$ 0.002	1.00	0.720 $\pm$ 0.007	1.45	0.635 $\pm$ 0.122	1.45
White blood cell dis.	194	0.718 $\pm$ 0.003	1.70	0.718 $\pm$ 0.003	1.70	0.720 $\pm$ 0.007	1.10	0.646 $\pm$ 0.115	1.10
Diabetes mellitus	434	0.715 $\pm$ 0.003	1.05	0.712 $\pm$ 0.004	1.00	0.717 $\pm$ 0.004	1.20	0.672 $\pm$ 0.039	1.20
Hypertension	847	0.714 $\pm$ 0.002	2.05	0.706 $\pm$ 0.004	2.55	0.715 $\pm$ 0.003	1.50	0.681 $\pm$ 0.022	1.50
Abnormal sputum	58	0.708 $\pm$ 0.036	3.50	0.731 $\pm$ 0.008	2.45	0.708 $\pm$ 0.026	1.10	0.632 $\pm$ 0.089	1.10
Other urinary disorders	156	0.708 $\pm$ 0.005	3.45	0.721 $\pm$ 0.002	1.45	0.714 $\pm$ 0.007	1.60	0.617 $\pm$ 0.101	1.60
Toxic solvent/alcohol effect	135	0.706 $\pm$ 0.008	2.45	0.704 $\pm$ 0.003	1.35	0.698 $\pm$ 0.016	1.35	0.652 $\pm$ 0.083	1.35
Aplastic anemia	78	0.701 $\pm$ 0.007	1.60	0.701 $\pm$ 0.003	1.70	0.685 $\pm$ 0.015	1.00	0.633 $\pm$ 0.084	1.00
Abdominal hernia	88	0.701 $\pm$ 0.027	1.00	0.712 $\pm$ 0.006	1.80	0.701 $\pm$ 0.014	1.65	0.613 $\pm$ 0.086	1.65
Malaise/fatigue	307	0.699 $\pm$ 0.002	3.45	0.700 $\pm$ 0.001	2.10	0.700 $\pm$ 0.003	1.75	0.660 $\pm$ 0.083	1.75
Mineral metabolism dis.	139	0.697 $\pm$ 0.006	1.85	0.700 $\pm$ 0.002	2.05	0.698 $\pm$ 0.006	2.45	0.585 $\pm$ 0.117	2.45
Sleep dis.	295	0.695 $\pm$ 0.005	1.70	0.692 $\pm$ 0.004	1.15	0.698 $\pm$ 0.002	1.30	0.601 $\pm$ 0.087	1.30
Kidney/ureter disorders	69	0.692 $\pm$ 0.011	2.80	0.688 $\pm$ 0.006	1.25	0.682 $\pm$ 0.017	1.40	0.638 $\pm$ 0.070	1.40
Other tests	97	0.688 $\pm$ 0.005	2.10	0.695 $\pm$ 0.004	3.00	0.688 $\pm$ 0.014	2.00	0.640 $\pm$ 0.047	2.00

Table S10. Concept-model comparison on INSPECT, part II. Continuation of the INSPECT concept-model comparison. AUROC values and LLM interpretability scores compare TRACE and CBM variants using refined or full concept banks. **Bold** indicates the best result in each row.

Finding	N	TRACE Refined set		TRACE Full set		CBM Refined set		CBM Full set	
		AUROC	LLM	AUROC	LLM	AUROC	LLM	AUROC	LLM
Cardiomegaly	228	0.687 ± 0.003	1.00	0.679 ± 0.004	2.05	0.686 ± 0.005	1.25	0.622 ± 0.066	1.25
Abnormal lung findings	957	0.686 ± 0.001	4.25	0.685 ± 0.001	3.70	0.687 ± 0.001	1.85	0.620 ± 0.089	1.85
Osteoarthritis	85	0.686 ± 0.015	1.00	0.692 ± 0.017	1.00	0.691 ± 0.011	1.00	0.587 ± 0.057	1.00
Urinary tract infection	90	0.686 ± 0.018	2.00	0.689 ± 0.009	1.80	0.689 ± 0.012	1.25	0.629 ± 0.049	1.25
Purpura/hemorrhagic dis.	134	0.680 ± 0.006	1.65	0.678 ± 0.003	1.75	0.678 ± 0.017	1.45	0.614 ± 0.104	1.45
Other liver dis.	212	0.677 ± 0.002	1.90	0.679 ± 0.001	2.00	0.681 ± 0.009	2.10	0.610 ± 0.109	2.10
Cerebrovascular dis.	108	0.676 ± 0.009	2.25	0.679 ± 0.004	1.00	0.670 ± 0.018	1.00	0.630 ± 0.032	1.00
Other aneurysm	55	0.671 ± 0.027	2.00	0.694 ± 0.007	1.05	0.668 ± 0.024	1.00	0.628 ± 0.079	1.00
Respiratory abnormalities	93	0.670 ± 0.006	3.35	0.675 ± 0.003	1.15	0.677 ± 0.013	1.30	0.638 ± 0.038	1.30
Muscle symptoms	74	0.669 ± 0.011	3.45	0.687 ± 0.006	1.30	0.673 ± 0.023	1.00	0.602 ± 0.087	1.00
Pancreatic diseases	58	0.668 ± 0.034	1.00	0.691 ± 0.010	1.00	0.672 ± 0.035	1.70	0.626 ± 0.098	1.70
Fever unknown origin	209	0.667 ± 0.003	3.50	0.665 ± 0.002	1.85	0.661 ± 0.008	2.10	0.582 ± 0.084	2.10
Alcohol-related dis.	86	0.666 ± 0.018	3.05	0.669 ± 0.010	2.65	0.675 ± 0.023	1.85	0.575 ± 0.108	1.85
Epilepsy/seizures	74	0.661 ± 0.015	1.00	0.658 ± 0.004	1.10	0.651 ± 0.024	1.05	0.609 ± 0.063	1.05
Acquired organ absence	163	0.660 ± 0.004	1.65	0.661 ± 0.004	1.75	0.652 ± 0.024	1.55	0.617 ± 0.078	1.55
Anxiety dis.	270	0.658 ± 0.006	1.75	0.662 ± 0.002	1.85	0.661 ± 0.014	1.00	0.574 ± 0.077	1.00
Functional digestive dis.	81	0.657 ± 0.010	1.90	0.677 ± 0.007	1.00	0.677 ± 0.017	1.70	0.634 ± 0.038	1.70
Venous embolism/thrombosis	235	0.657 ± 0.005	2.45	0.665 ± 0.002	1.15	0.656 ± 0.012	1.00	0.614 ± 0.094	1.00
Cardiac conduction dis.	981	0.653 ± 0.001	1.25	0.654 ± 0.002	1.55	0.659 ± 0.001	1.30	0.614 ± 0.057	1.30
Other lung dis.	144	0.649 ± 0.004	3.75	0.644 ± 0.003	3.85	0.651 ± 0.007	2.20	0.597 ± 0.050	2.20
MSK nonspecific findings	53	0.643 ± 0.019	3.15	0.642 ± 0.014	1.75	0.640 ± 0.024	1.15	0.600 ± 0.041	1.15
Abdominal pain	274	0.637 ± 0.005	2.05	0.638 ± 0.005	1.75	0.634 ± 0.016	1.30	0.593 ± 0.050	1.30
Nonspecific chest pain	701	0.636 ± 0.003	2.15	0.632 ± 0.007	2.80	0.645 ± 0.003	1.50	0.541 ± 0.066	1.50
Esophageal diseases	342	0.634 ± 0.003	1.15	0.636 ± 0.002	1.10	0.637 ± 0.003	1.25	0.587 ± 0.059	1.25
Substance addiction/dis.	81	0.632 ± 0.026	1.30	0.632 ± 0.022	1.00	0.631 ± 0.027	1.00	0.565 ± 0.084	1.00
Rheumatic valve dis.	62	0.632 ± 0.014	2.45	0.635 ± 0.004	2.15	0.643 ± 0.009	2.10	0.584 ± 0.029	2.10
Abdomen/pelvis symptoms	65	0.629 ± 0.013	2.15	0.622 ± 0.017	1.65	0.623 ± 0.016	3.25	0.568 ± 0.075	3.25
Cardiac/circulatory anomalies	56	0.621 ± 0.033	1.70	0.632 ± 0.012	1.90	0.604 ± 0.036	1.40	0.591 ± 0.035	1.40
Cellulitis/abscess	55	0.618 ± 0.009	1.65	0.628 ± 0.003	1.80	0.623 ± 0.019	1.30	0.576 ± 0.056	1.30
Other upper resp. dis.	50	0.616 ± 0.011	2.60	0.616 ± 0.006	1.70	0.612 ± 0.016	1.30	0.585 ± 0.044	1.30
Injury	58	0.615 ± 0.024	3.05	0.629 ± 0.027	1.00	0.635 ± 0.024	1.75	0.540 ± 0.042	1.75
Inflammatory/toxic neuropathy	51	0.614 ± 0.016	1.95	0.626 ± 0.005	1.40	0.598 ± 0.034	1.85	0.605 ± 0.031	1.85
Nausea/vomiting	188	0.614 ± 0.005	1.90	0.617 ± 0.003	1.85	0.619 ± 0.007	1.55	0.587 ± 0.052	1.55
Mood dis.	242	0.612 ± 0.004	1.00	0.613 ± 0.003	1.00	0.614 ± 0.005	1.00	0.571 ± 0.041	1.00
Pulmonary heart dis.	459	0.601 ± 0.005	1.85	0.603 ± 0.006	1.85	0.603 ± 0.005	1.25	0.570 ± 0.049	1.25
Other soft tissue dis.	108	0.598 ± 0.015	1.75	0.573 ± 0.006	2.50	0.595 ± 0.012	1.85	0.529 ± 0.024	1.85
Other hematologic dis.	171	0.597 ± 0.007	2.00	0.603 ± 0.002	1.65	0.601 ± 0.009	1.40	0.570 ± 0.054	1.40
Joint pain	108	0.592 ± 0.028	1.85	0.594 ± 0.024	1.40	0.595 ± 0.019	1.10	0.519 ± 0.025	1.10
Asthma	189	0.588 ± 0.008	1.95	0.586 ± 0.009	2.00	0.588 ± 0.006	1.20	0.528 ± 0.020	1.20
Skin symptoms	90	0.586 ± 0.012	1.95	0.596 ± 0.014	1.70	0.600 ± 0.019	1.55	0.562 ± 0.025	1.55
Limb MSK symptoms	121	0.586 ± 0.015	1.45	0.567 ± 0.007	2.05	0.584 ± 0.011	1.45	0.529 ± 0.027	1.45
Breast cancer	90	0.583 ± 0.012	2.70	0.581 ± 0.006	2.65	0.586 ± 0.020	1.30	0.542 ± 0.046	1.30
Back pain	124	0.578 ± 0.010	2.05	0.574 ± 0.011	1.40	0.571 ± 0.013	1.25	0.527 ± 0.043	1.25
Nonspecific blood findings	186	0.575 ± 0.012	2.40	0.574 ± 0.010	2.10	0.585 ± 0.016	1.60	0.526 ± 0.041	1.60
Syncope/collapse	98	0.574 ± 0.033	1.20	0.586 ± 0.035	1.70	0.597 ± 0.020	1.75	0.485 ± 0.049	1.75
Tobacco use dis.	125	0.570 ± 0.011	2.00	0.571 ± 0.010	1.95	0.574 ± 0.012	1.10	0.535 ± 0.026	1.10
Other headache syndromes	50	0.569 ± 0.036	1.50	0.604 ± 0.037	1.60	0.580 ± 0.035	1.25	0.527 ± 0.037	1.25
Limb pain	99	0.553 ± 0.015	2.00	0.558 ± 0.023	2.00	0.563 ± 0.025	1.00	0.515 ± 0.030	1.00
Vestibular dis.	94	0.542 ± 0.036	1.00	0.532 ± 0.035	1.00	0.554 ± 0.016	1.00	0.468 ± 0.061	1.00
Leukemia	62	0.531 ± 0.010	1.90	0.533 ± 0.004	2.35	0.524 ± 0.025	1.55	0.528 ± 0.030	1.55

Table S11. Concept-model comparison on CT-RATE. AUROC values and LLM interpretability scores compare TRACE and CBM variants using refined or full concept banks. **Bold** indicates the best result in each row.

Finding	N	TRACE Refined set		TRACE Full set		CBM Refined set		CBM Full set	
		AUROC	LLM	AUROC	LLM	AUROC	LLM	AUROC	LLM
Pleural effusion	336	0.949 $\pm$ 0.003	3.80	0.944 $\pm$ 0.003	3.75	0.950 $\pm$ 0.002	1.85	0.935 $\pm$ 0.014	1.85
Cardiomegaly	273	0.898 $\pm$ 0.005	2.05	0.895 $\pm$ 0.004	2.45	0.903 $\pm$ 0.003	2.35	0.878 $\pm$ 0.016	2.35
Arterial wall calcification	733	0.883 $\pm$ 0.004	4.60	0.885 $\pm$ 0.002	4.00	0.884 $\pm$ 0.002	1.10	0.857 $\pm$ 0.039	1.10
Consolidation	490	0.878 $\pm$ 0.007	3.85	0.878 $\pm$ 0.007	3.15	0.880 $\pm$ 0.005	1.95	0.846 $\pm$ 0.038	1.95
Coronary artery calc.	643	0.875 $\pm$ 0.005	4.40	0.874 $\pm$ 0.005	4.00	0.875 $\pm$ 0.003	1.35	0.852 $\pm$ 0.025	1.35
Interlobular septal thick.	206	0.857 $\pm$ 0.017	2.85	0.872 $\pm$ 0.002	1.80	0.867 $\pm$ 0.008	1.55	0.842 $\pm$ 0.039	1.55
Pericardial effusion	205	0.849 $\pm$ 0.003	1.95	0.843 $\pm$ 0.003	1.50	0.841 $\pm$ 0.005	2.55	0.793 $\pm$ 0.080	2.55
Lung opacity	971	0.839 $\pm$ 0.005	4.20	0.838 $\pm$ 0.009	3.80	0.841 $\pm$ 0.003	2.20	0.802 $\pm$ 0.045	2.20
Mosaic attenuation	213	0.817 $\pm$ 0.008	1.90	0.808 $\pm$ 0.013	2.05	0.831 $\pm$ 0.007	1.80	0.753 $\pm$ 0.083	1.80
Emphysema	496	0.786 $\pm$ 0.004	1.45	0.783 $\pm$ 0.007	2.55	0.785 $\pm$ 0.002	1.25	0.751 $\pm$ 0.028	1.25
Peribronchial thick.	312	0.780 $\pm$ 0.006	1.90	0.788 $\pm$ 0.006	2.15	0.788 $\pm$ 0.008	2.30	0.728 $\pm$ 0.087	2.30
Bronchiectasis	267	0.779 $\pm$ 0.008	3.25	0.778 $\pm$ 0.012	3.75	0.787 $\pm$ 0.007	1.35	0.716 $\pm$ 0.048	1.35
Medical material	277	0.777 $\pm$ 0.002	3.70	0.774 $\pm$ 0.005	4.25	0.782 $\pm$ 0.007	2.25	0.753 $\pm$ 0.018	2.25
Atelectasis	600	0.741 $\pm$ 0.006	4.70	0.745 $\pm$ 0.005	3.40	0.744 $\pm$ 0.004	2.50	0.721 $\pm$ 0.022	2.50
Hiatal hernia	341	0.736 $\pm$ 0.015	1.00	0.750 $\pm$ 0.006	1.00	0.745 $\pm$ 0.010	1.15	0.669 $\pm$ 0.075	1.15
Lymphadenopathy	693	0.723 $\pm$ 0.004	1.70	0.724 $\pm$ 0.005	1.55	0.726 $\pm$ 0.002	1.40	0.714 $\pm$ 0.009	1.40
Pulm. fibrotic sequela	673	0.699 $\pm$ 0.009	3.40	0.700 $\pm$ 0.008	3.95	0.707 $\pm$ 0.005	1.35	0.663 $\pm$ 0.028	1.35
Lung nodule	1123	0.689 $\pm$ 0.005	2.70	0.688 $\pm$ 0.012	2.00	0.691 $\pm$ 0.002	1.70	0.661 $\pm$ 0.041	1.70

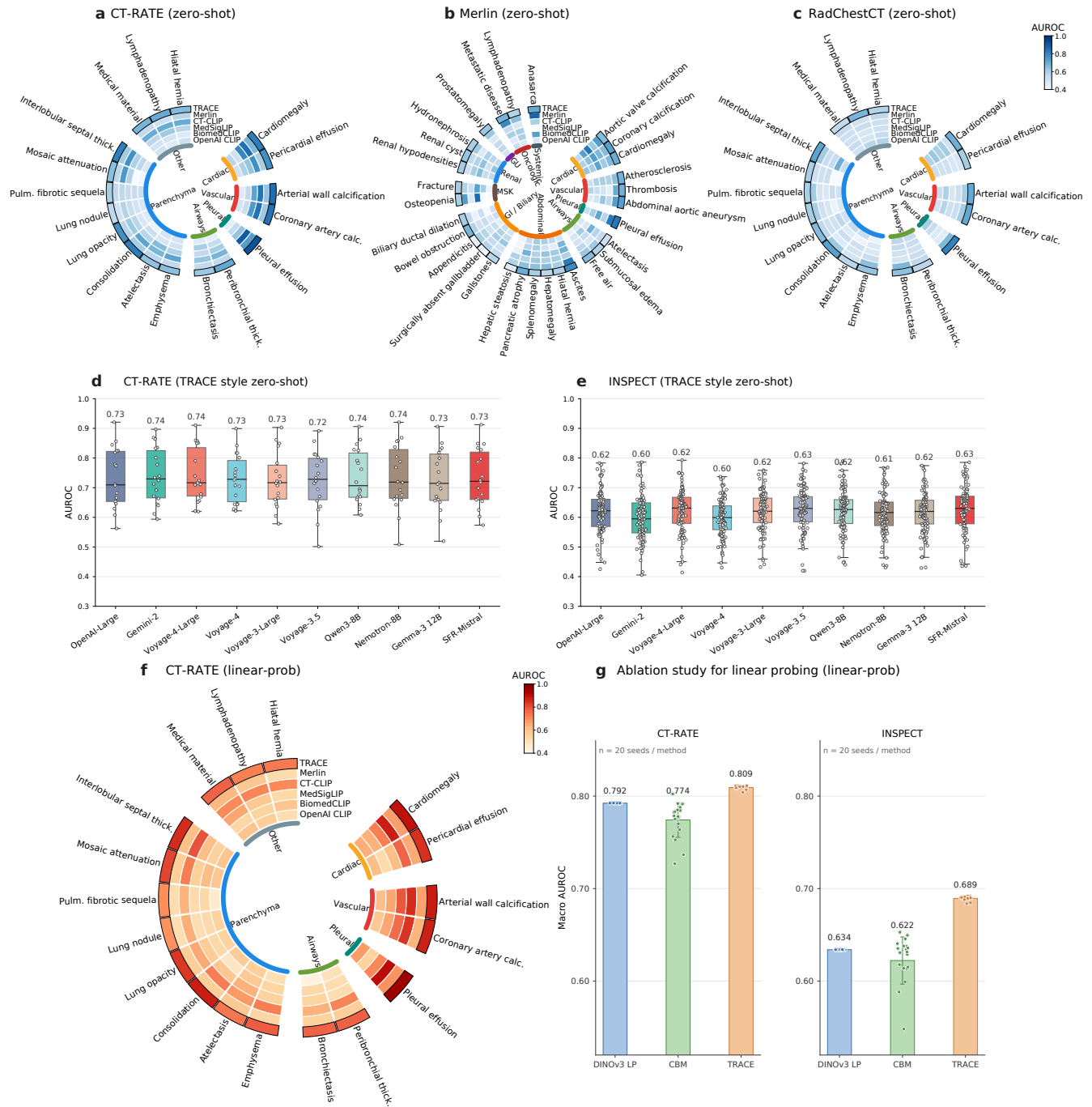


Figure S1. Robustness of TRACE under zero-shot and linear-probing evaluation. **a-c**, Zero-shot disease-level AUROC profiles on CT-RATE, Merlin and RadChestCT, respectively, using OpenAI text-embedding-3-large text embeddings. TRACE is compared with representative vision-language and medical foundation-model baselines. Findings are organized by anatomical or clinical system, showing that the performance of TRACE is broadly distributed across disease groups rather than driven by a small subset of labels. **d,e**, Zero-shot AUROC distributions on CT-RATE and INSPECT under alternative concept-bank construction strategies. Each point denotes one disease label, and boxes summarize the interquartile range across labels, indicating that TRACE remains stable across different concept-selection schemes and datasets. **f**, Linear-probing disease-level AUROC on CT-RATE using OpenAI text-embedding-3-large text embeddings, showing consistent label-wise discrimination under a standard linear-evaluation setting. **g**, Linear-probing ablation on CT-RATE and INSPECT. Bars denote the mean macro AUROC over 20 random seeds, with individual seed results overlaid, demonstrating the contribution of TRACE beyond the DINOv3 linear probe and the conventional concept bottleneck model.

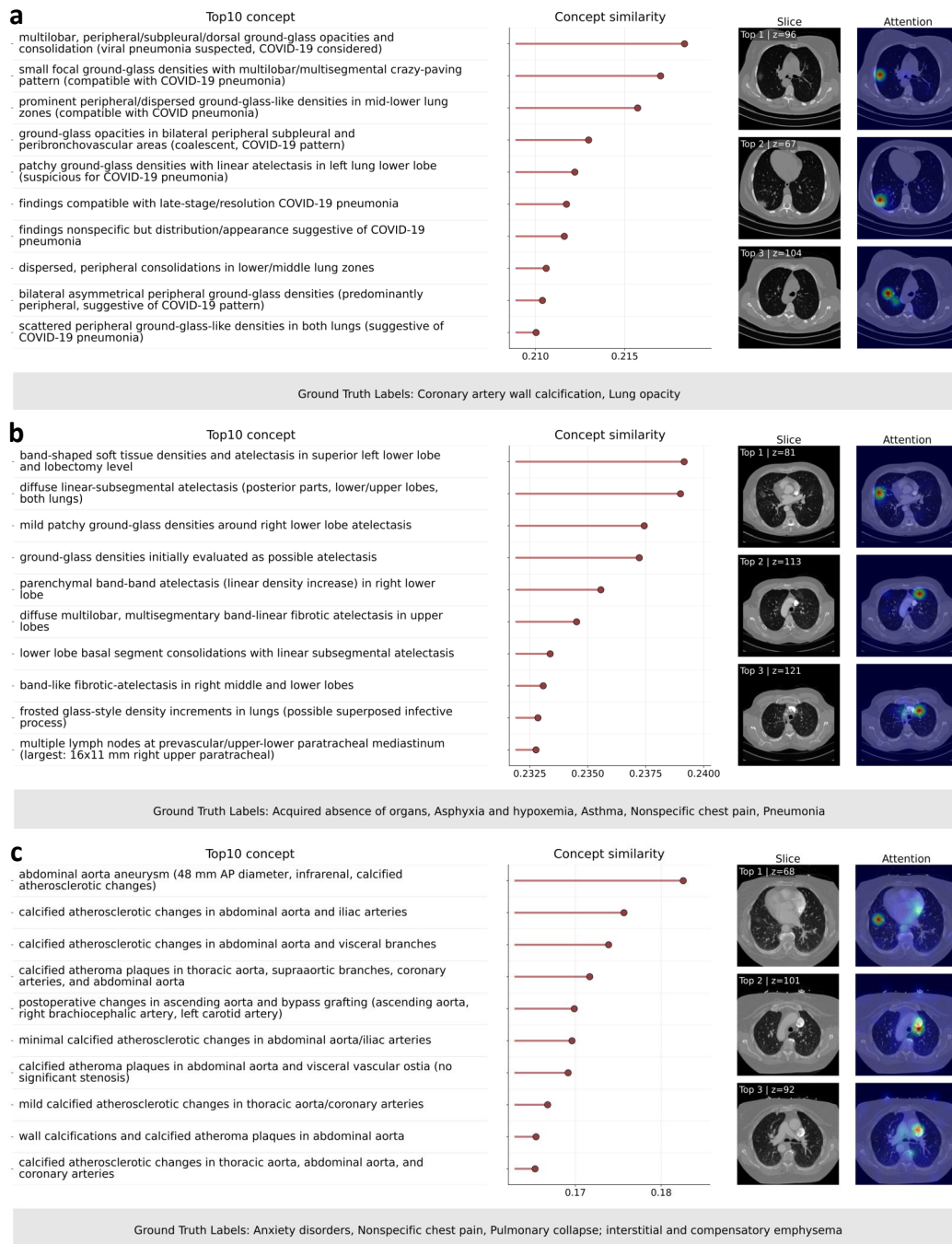


Figure S2. Representative case-level TRACE explanations. **a**, Example in which top-ranked concepts are dominated by device- and procedure-related findings, including biliary stents and postoperative change. **b**, Example emphasizing parenchymal abnormalities, including atelectatic and fibrotic changes. **c**, Example highlighting pleural or cardiopulmonary concepts. For each case, TRACE ranks the concepts that most strongly support the prediction, illustrating how case-level outputs can be inspected in clinically meaningful language.

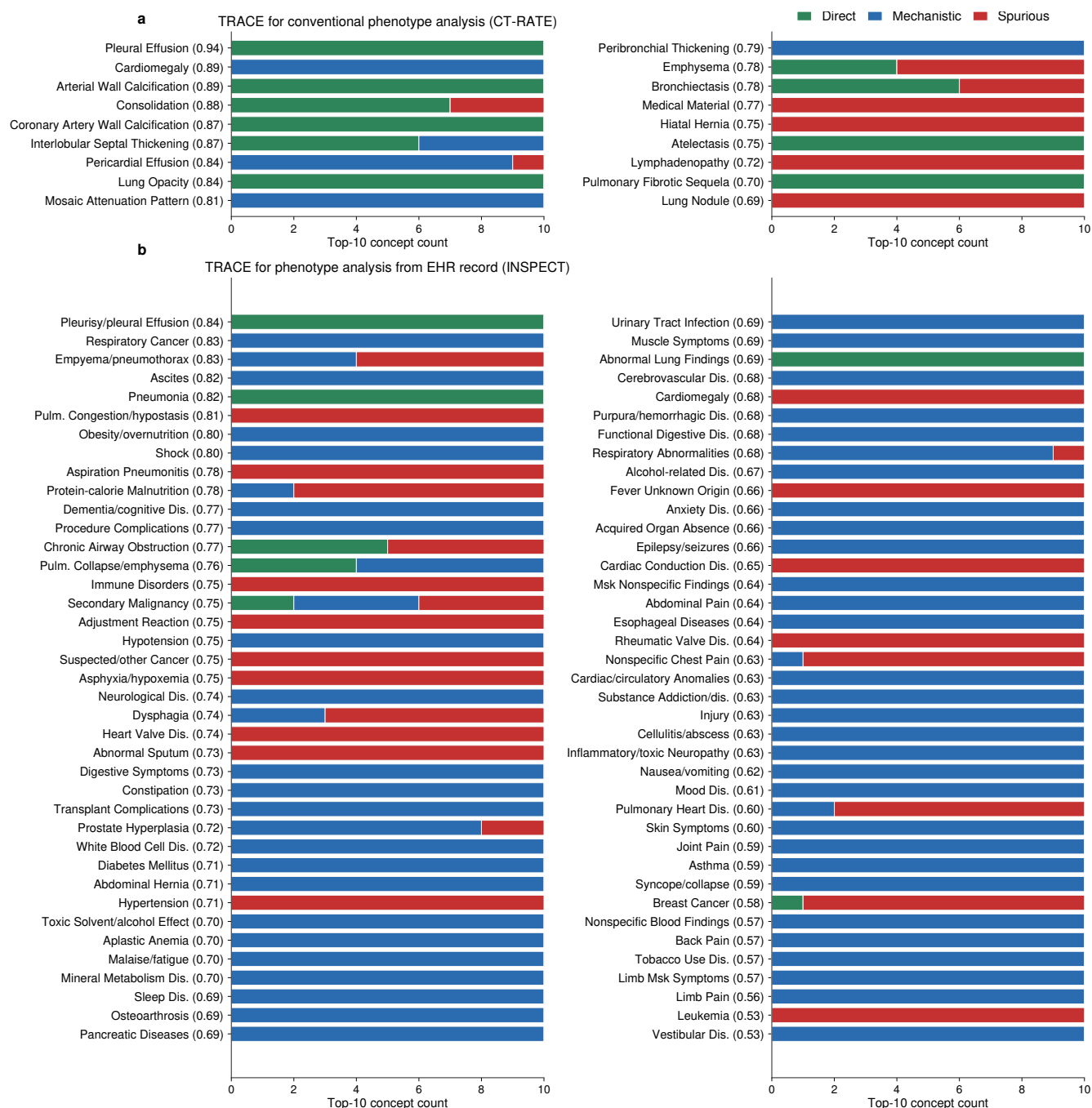


Figure S3. Manual audit of top-ranked TRACE concepts. **a**, Audit composition of the top-10 concepts for CT-RATE radiological phenotypes. Directly image-defined findings, such as pleural effusion, cardiomegaly and arterial calcification, are dominated by expected concepts. **b**, Audit composition for high-performing INSPECT phenotypes. INSPECT includes EHR-derived and opportunistic labels, producing a broader mixture of expected concepts, candidate associations, confounders and artefacts.



Figure S4. Clinical audit of class-wise top-10 TRACE concepts on CT-RATE. For each CT-RATE phenotype, expert clinicians annotated the ten highest-ranked TRACE concepts. The labels distinguish expected diagnostic evidence from candidate associations, confounders and artefacts, enabling class-wise inspection of whether the model relies on clinically plausible concepts.



Figure S5. Clinical audit of class-wise top-10 TRACE concepts on INSPECT, part I. Expert clinician annotations are shown for the ten highest-ranked TRACE concepts for the first group of INSPECT phenotypes. The audit categories indicate whether each concept represents expected evidence, a candidate disease association, a confounder or an artefact.

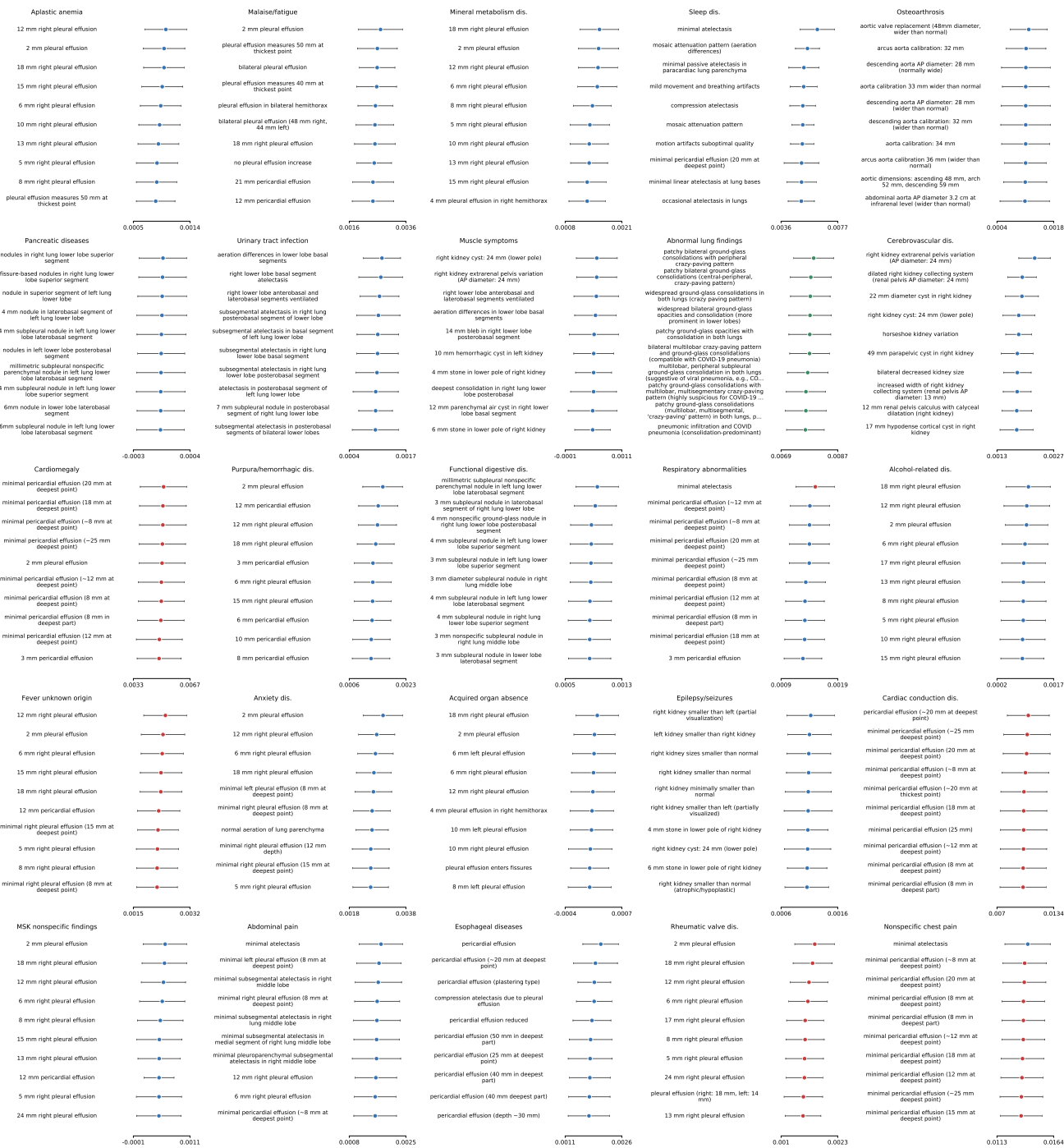


Figure S6. Clinical audit of class-wise top-10 TRACE concepts on INSPECT, part II. Continuation of the clinician audit for INSPECT. The top-ranked concepts for each phenotype are annotated using the same four-category scheme, supporting systematic review of model reasoning across EHR-derived labels.

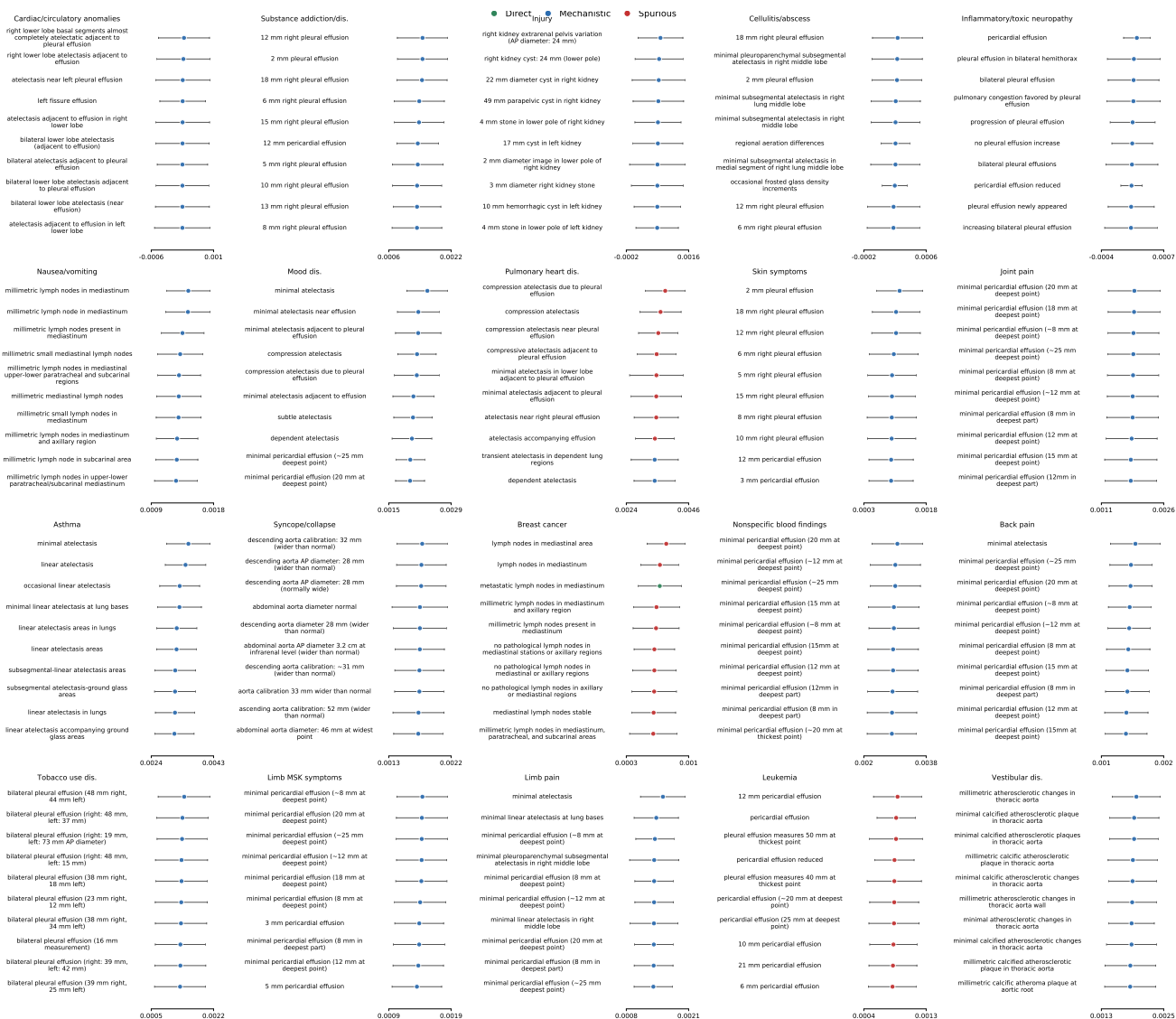


Figure S7. Clinical audit of class-wise top-10 TRACE concepts on INSPECT, part III. Final set of INSPECT clinician audits. For each phenotype, the top ten TRACE concepts are shown with audit labels, allowing expected, novel, confounding and artefactual concept patterns to be compared across disease systems.

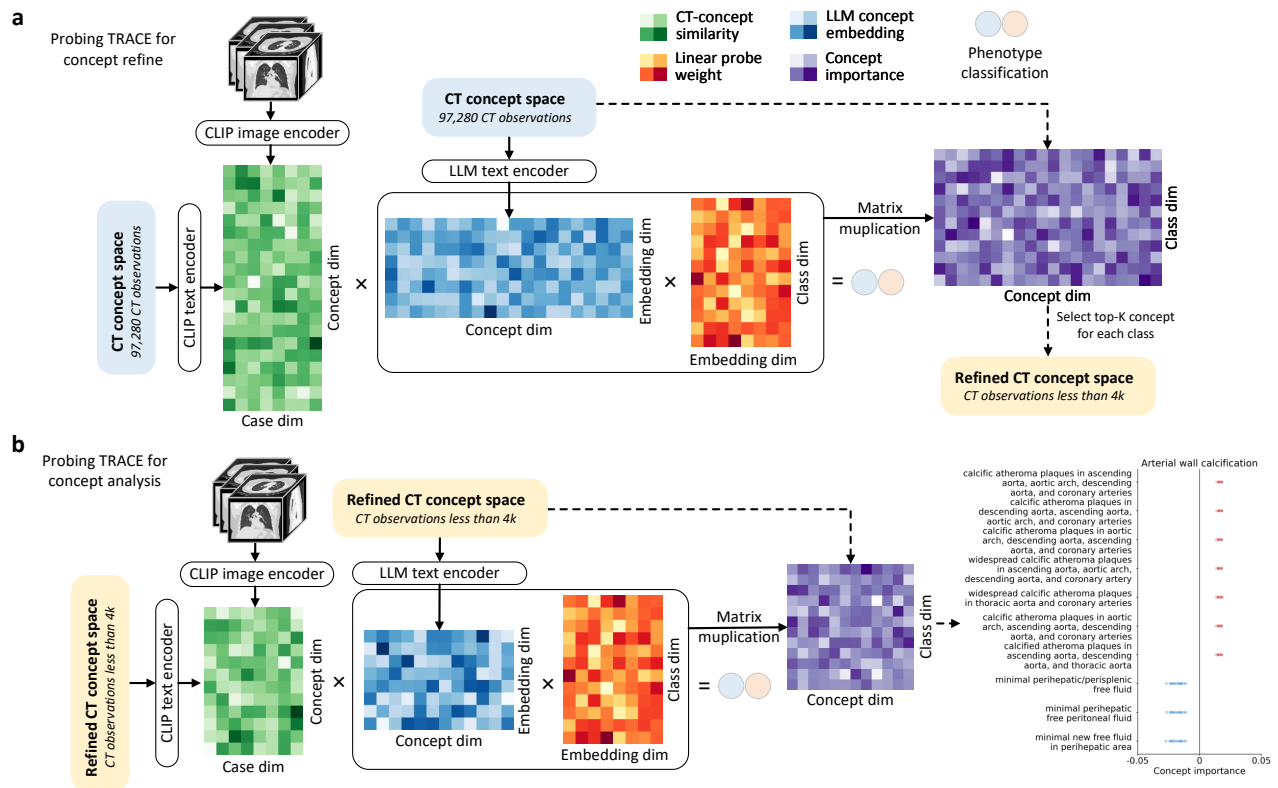


Figure S8. TRACE workflow for concept refinement and concept analysis. **a**, Concept refinement combines CT-concept similarity scores, LLM-derived concept embeddings and linear-probe weights to obtain class-specific concept-importance scores. Top-ranked concepts are selected to form a compact refined concept space. **b**, The refined space is reused for downstream concept analysis, producing phenotype-specific ranked concept profiles with fewer and more relevant concepts.

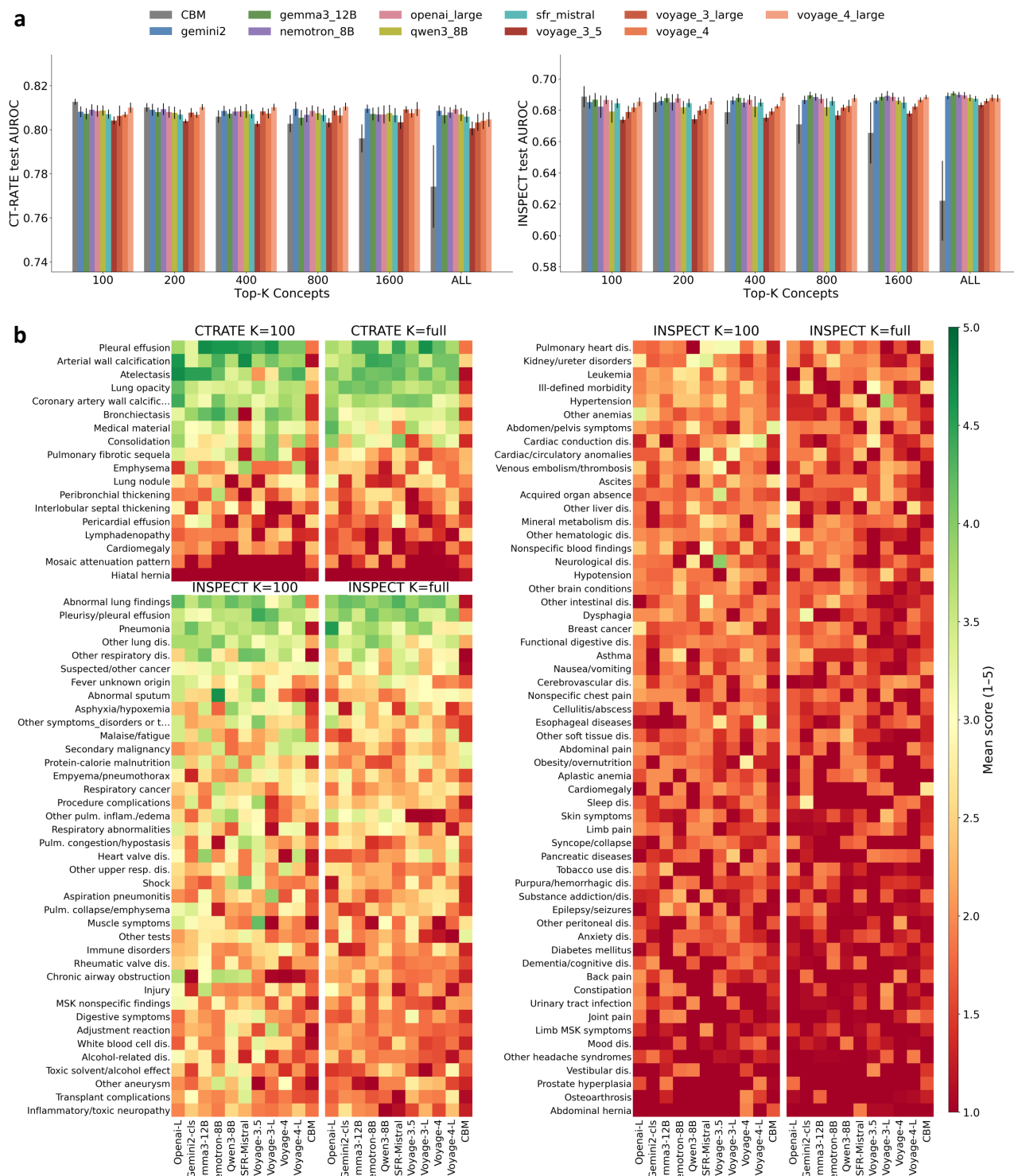


Figure S9. Sensitivity to concept-bank size and LLM text encoder. **a**, CT-RATE and INSPECT AUROCs are compared across top- K concept banks and full concept sets generated by different LLM text encoders. Stable performance across a wide range of K values indicates that TRACE is not driven by a single arbitrary concept-bank size. **b**, Mean audit scores for CT-RATE and INSPECT explanations under $K=100$ and full-concept settings, showing how explanation quality varies across phenotypes and text encoders.

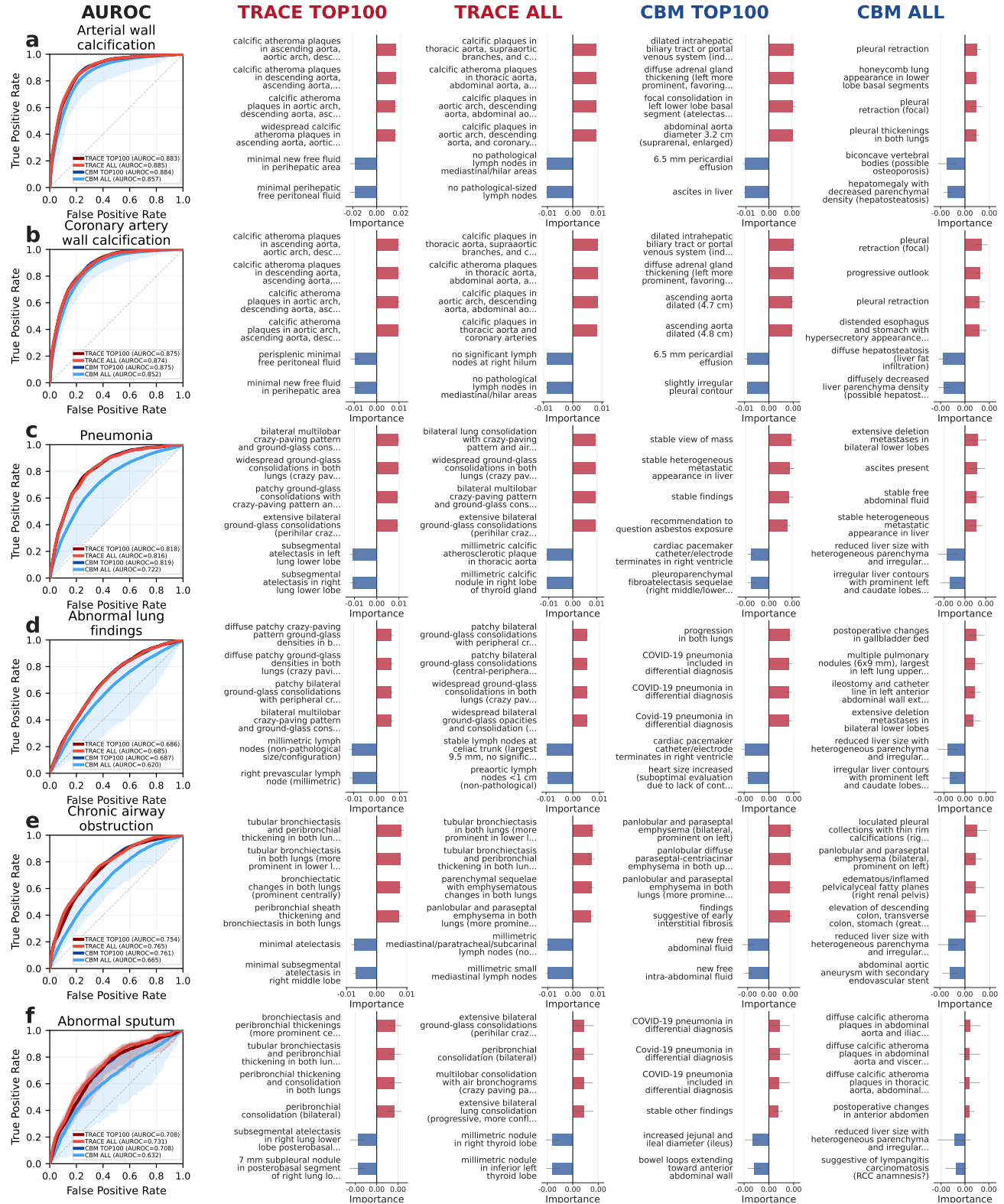


Figure S10. TRACE and CBM explanations across representative phenotypes. a-f, AUROC curves and concept profiles visualization. For each phenotype, the left column reports predictive performance, whereas the remaining columns compare TRACE and CBM explanations under top-100 and all-concept settings. TRACE more consistently ranks phenotype-specific and clinically aligned concepts, whereas CBM profiles more often include diffuse or less directly related concepts.

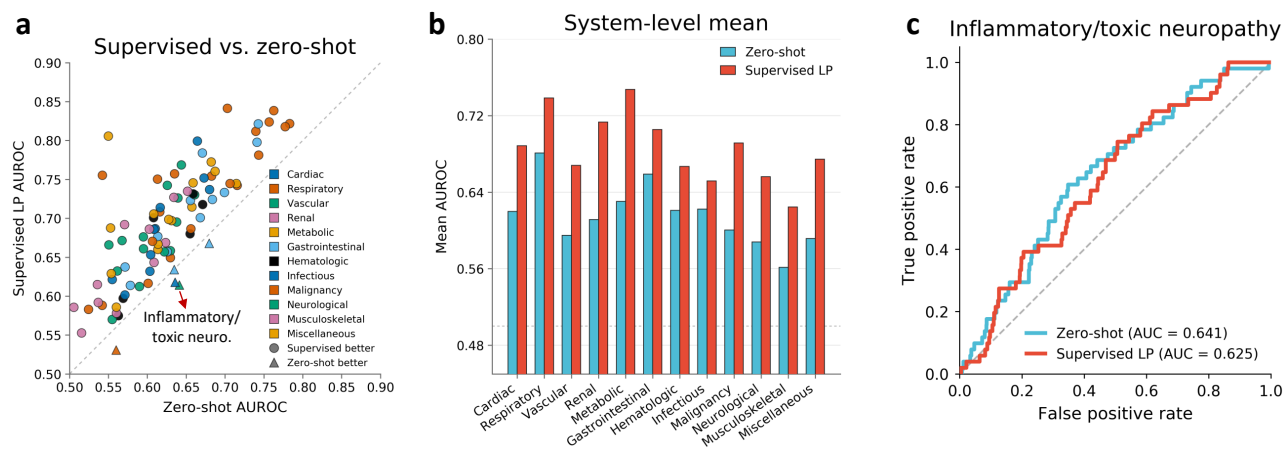


Figure S11. Supervised and zero-shot phenotype prediction on INSPECT. **a**, Class-level comparison between supervised linear probing and zero-shot prediction. Each point represents an INSPECT phenotype, colours denote disease systems and the diagonal indicates equal performance. **b**, Mean AUROC by disease system for zero-shot prediction and linear probing, showing the effect of supervised calibration. **c**, ROC curve for inflammatory or toxic neuropathy, an example in which zero-shot prediction remains competitive with supervised linear probing.